PHY354 Course Notes

Hannah Guo

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1 Lagrangian Mechanics and the Principle of least Action

Example 1 Simple Spring

Consider simple mass on spring problem, we have

$$F = -kx$$

Describe in Lagrangian with the following steps:

$$L = T - V \tag{1}$$

$$T = \frac{1}{2}mv^2$$

$$V = \frac{1}{2}kx^2$$

We have Euler-Lagrange Equations:

$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{x}}) = \frac{\partial l}{\partial x} \tag{2}$$

We eventually get: $m\ddot{x} = -kx$

Example 2 General Potential

We have Euler-Lagrange Equations as:

$$m\ddot{x} = -\frac{\partial V}{\partial x}$$

In 3D, we have

$$L = \frac{1}{2}m\sum_i \dot{x_i} - V(x_i) \tag{3}$$

$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{x_i}}) = \frac{\partial L}{\partial x_i}$$

$$m\ddot{x_i} = \frac{\partial L}{\partial x_i}$$

Example 3 Different Coordinates

Consider the following system:

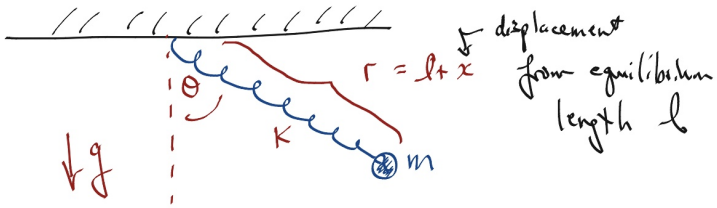


Figure 1: Mass on Spring

We can choose coordinates x and θ. Thus we have the following expressions:

$$T = \frac{1}{2}m\vec{v}^2 = \frac{1}{2}(\dot{x}^2 + (l+x)^2\dot{\theta}^2) \tag{4}$$

$$V = -mg(l+x)\cos\theta + \frac{1}{2}kx^2 \tag{5}$$

Therefore, the E-L Equations are:

$$m\ddot{x} = m(l+x)\dot{\theta}^2 + mg\cos\theta - kx \tag{6}$$

$$m(l+x)\ddot{\theta} + 2m\dot{x}\dot{\theta} = -mg\sin\theta \tag{7}$$

(12) is the radial component of $\vec{F} = m\vec{a}$, which includes the centripetal acceleration.
(13) is $\vec{\tau} = \frac{d\vec{M}}{dt}$

Principle of least Action

To specify the position of a particle in 3 dimensions, we need 3 coordinates q_i. If we have N particles, we will need 3N coordinates to describe the state of the system.
Back to Example 3, the generalized coordinates are $q_1(t) = x(t)$ and $q_2(t) = \theta(t)$.
The main problem in mechanics is:

$$\text{Given } \vec{r_i}(t_1),\ \dot{\vec{r_i}}(t_1)$$
$$\text{Find } \vec{r_i}(t_2),\ \dot{\vec{r_i}}(t_2)$$

Example 4 1D Particle

Imagine we have a particle with some external force acting on it. Let's assume we know that at time $t = t_1$, the particle is at $q(t_1) = q_1$. Then, let's ask the question "what path did the particle take to go from q_0 to q_1?"
To each path, assign a number called the Action:

$$S = \int_{t_0}^{t_1} L(q(t), \dot{q}(t), t)dt \tag{8}$$

S is the function of the path defined by q(t). Our claim is the the particle will take a path which minimize the Action S. Taking the derivative to find the minimum of the action and we arrive with the Euler-Lagrange equations.

Comments

The Lagrangian for a system is not unique:

- Can multiply L by a constant $\rightarrow \alpha L$
- Can add the total time derivative of a function $\rightarrow L + \frac{df}{dt}$ for any $f(q_i, t)$

In general, if we have the integral

$$I = \int_{x_1}^{x_2} F(y(x_1), \dot{y}(x_1))dx \tag{9}$$

then I is extremized y the path y(x) which satisfies

$$\frac{d}{dx}(\frac{\partial F}{\partial \dot{y}(x)}) = \frac{dF}{dy} \tag{10}$$

Example 5 Brachistochrone

By conservation of energy, we have

$$\frac{1}{2}mv^2 = mgx \tag{11}$$

Therefore $v = \sqrt{2gx}$.
Noting the distance along the curve satisfies $ds^2 = dx^2 + dy^2$, we can express the time $T(y)$ taken for the particle to descend along the curve $y = y(x)$ as a functional:

$$T(y) = \int_A^B dt = \int_A^B \frac{ds}{v} = \int_0^h \frac{\sqrt{1 + (\dot{y}^2)}}{\sqrt{2gx}} \tag{12}$$

where $y(0) = 0$, $y(h) = a$.
Observe this expression is a functional satisfying the Euler-Lagrange equation. Therefore:

$$\frac{d}{dx}\left(\frac{y'}{\sqrt{2gx\left(1 + (y')^2\right)}}\right) = 0, \quad y(0) = 0, y(h) = c \tag{13}$$

Integrating this, we get

$$\frac{y'}{\sqrt{2gx\left(1 + (y')^2\right)}} = c \tag{14}$$

Rearranging for y' and solve for y:

$$y' = \frac{dy}{dx} = \frac{\sqrt{x}}{\sqrt{\alpha - x}}, \quad \text{with } \alpha = \frac{1}{2gc^2} \tag{15}$$

We can integrate this equation using the substitution $x = a\sin^2\theta$ to obtain

$$y = \int \frac{\sqrt{x}}{\sqrt{\alpha - x}}\,dx = \int \frac{\sin\theta}{\cos\theta} 2\alpha \sin\theta\cos\theta\,d\theta = \int \alpha(1 - \cos 2\theta)\,d\theta = \frac{\alpha}{2}(2\theta - \sin 2\theta) + k \tag{16}$$

Substituting back for x, and using y(0) = 0 to set k = 0, we obtain:

$$y(x) = \alpha \sin^{-1}(\sqrt{\frac{x}{\alpha}}) - \sqrt{x}\sqrt{\alpha - x} \tag{17}$$

This curve is a cycloid.

Constrained System

Example 6 Simple Pendulum

Consider the following system:

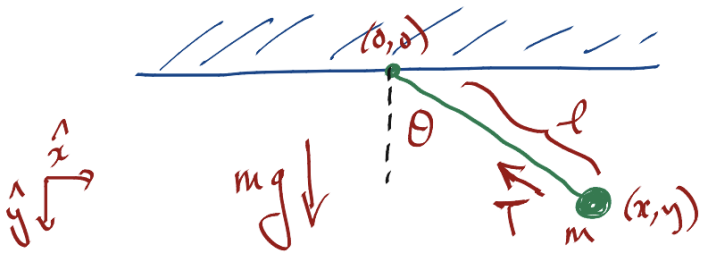


Figure 2: Simple Pendulum

Newtonian approach:

$$m\ddot{x} = -T\cos\theta = -T\frac{x}{l} \tag{18}$$

$$m\ddot{y} = mg - T\sin\theta = mg - T\frac{y}{l} \tag{19}$$

We have the constraint $x^2 + y^2 = l^2$.
Thus we solve for:

$$\ddot{\theta} = -\frac{g}{l}\sin\theta \tag{20}$$

$$T = ml\dot{\theta}^2 + mg\cos\theta \tag{21}$$

Lagrangian approach:
Identify the unconstrained coordinate θ and constrained coordinate $x^2 + y^2 = l^2$
The energies are:

$$T = \frac{1}{2}ml^2\dot{\theta}^2 \tag{22}$$

$$V = -mgl\cos\theta \tag{23}$$

Apply Euler-Lagrange and we arrive with:

$$\ddot{\theta} = -\frac{g}{l}\sin\theta \tag{24}$$

Example 7 Spherical Pendulum

Consider the following system:

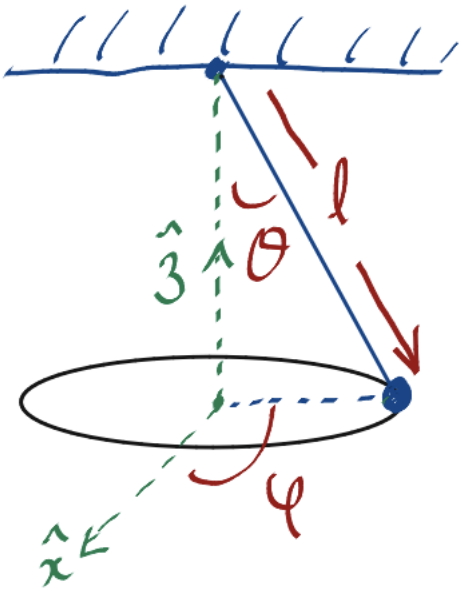


Figure 3: Spherical pendulum

The generalized coordinates in this case are (θ, φ). The trick is to write $\vec{v}^2$ in spherical polar coordinates.

$$x = r\sin\theta\cos\varphi \tag{25}$$
$$y = r\sin\theta\sin\varphi \tag{26}$$
$$z = r\cos\varphi \tag{27}$$

For spherical pendulum, we have $\dot{l} = 0$, so the Lagrangian is:

$$L = \frac{1}{2}ml^2(\dot{\theta}^2 + \dot{\varphi}^2 \\ sin^2\theta) + mgl\cos\theta \tag{28}$$

Thus we have two E-L equations:

$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{\theta}}) = \frac{\partial L}{\partial \theta} \Rightarrow ml^2\ddot{\theta} - ml^2\sin\theta\cos\theta\dot{\varphi}^2 + mgl\sin\theta = 0 \tag{29}$$
$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{\varphi}}) = 0 \Rightarrow \frac{d}{dt}(ml^2\sin^2\theta\dot{\varphi}) = 0 \tag{30}$$

Notice $ml^2\sin^2\theta\dot{\varphi}$ is the angular momentum about the $\hat{z}$ axis.

Example 8 2D Pendulum on Rotating Disk

Consider the following system:

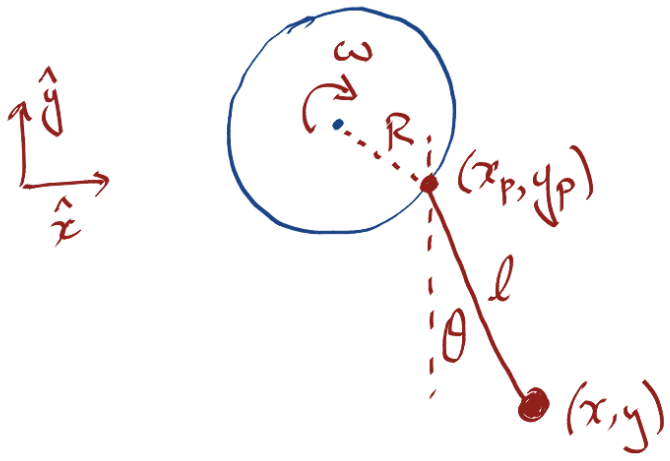


Figure 4: Rotating Pendulum

The only generalized coordinate is θ
To work out $\vec{v}^2$ of m, we need the coordinates of the mass. Let's establish the coordinates of the pivot first:

$$x_p = R\cos\omega t \tag{31}$$
$$y_p = -R\sin\omega t \tag{32}$$

Therefore the coordinates of the mass are:

$$x = x_p + l\sin\theta = R\cos\omega t + l\sin\theta$$
$$y = y_p - l\cos\theta = -R\sin\omega t - l\cos\theta$$

We now determine the velocities as:

$$\dot{x} = -\omega R\sin\omega t + l\cos\theta\dot{\theta}$$
$$\dot{y} = -\omega R\cos\omega + l\sin\theta\dot{\theta}$$
$$\vec{v}^2 = \dot{x}^2 + \dot{y}^2 = R^2\omega^2 - 2l\dot{\theta}R\omega\cos(\theta+\omega t) + l^2\dot{\theta}^2$$

Therefore, the Lagrangian of the system is:

$$l = \frac{1}{2}m(R^2\omega^2 - 2l\dot{\theta}R\omega\cos(\theta+\omega t) + l^2\dot{\theta}^2) + mg(R\sin\omega t + l\cos\theta) \tag{33}$$

Apply the E-L relations and we arrive with:

$$\ddot{\theta} + \frac{g}{l}\sin\theta = \frac{R\omega^2}{l}\cos\theta + \omega t$$

Example 9 Double Pendulum

Consider the following system:

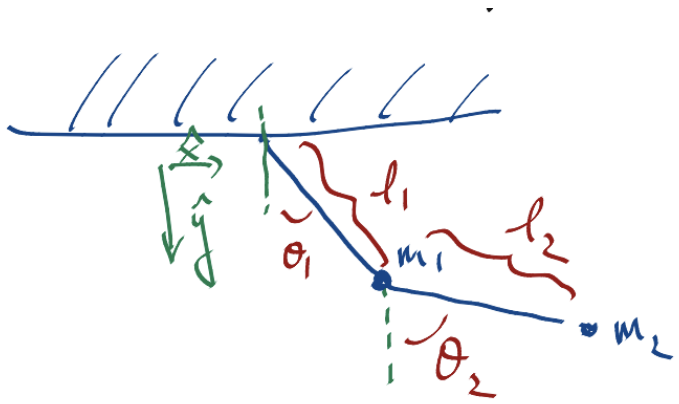


Figure 5: Double Pendulum

We need the positions of the two masses:

$$\begin{aligned} x_1 &= l_1 \sin\theta_1 \\ y_1 &= l_1 \cos\theta_1 \end{aligned} \tag{34}$$

$$\begin{aligned} x_2 &= x_1 + l_2 \sin\theta_2 = l_1 \sin\theta_1 + l_2 \sin\theta_2 \\ y_2 &= y_1 + l_2 \cos\theta_2 = l_1 \cos\theta_1 + l_2 \cos\theta_2 \end{aligned} \tag{35}$$

Therefore the kinetic energies are:

$$\begin{aligned} T_1 &= \frac{1}{2} m_1 (\dot{x_1}^2 + \dot{y_1}^2) = \frac{1}{2} m_1 l_1^2 \dot{\theta_1}^2 \\ T_2 &= \frac{1}{2} m_2 (\dot{x_2}^2 + \dot{y_2}^2) = \frac{1}{2} m_2 (l_1^2 \dot{\theta_1}^2 + l_2^2 \dot{\theta_2}^2 + 2 l_1 l_2 \dot{\theta_1} \dot{\theta_2} \cos(\theta_1 - \theta_2)) \end{aligned}$$

The potential energy of the system is:

$$V = -m_1 g l_1 \cos\theta_1 - m_2 g (l_1 \cos\theta_1 + l_2 \cos\theta_2) \tag{36}$$

The Lagrangian of the whole system is $T_1 + T_2 - V$.

Exercise: Find the E-L equations for $\theta_1$and θ_2 (Example of a Chaotic System)

Cyclic Coordinates, Symmetries, & Conservation Laws

Consider the the generalized coordinate q_i (but not $\dot{q}_i$), then q_i is said to be cyclic if we have

$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{q}_i}) = 0 \Rightarrow \frac{\partial L}{\partial \dot{q}_i} \text{ is constant (Independent of time).} \tag{37}$$

Example 10 Free Particle 1D

$$L = \sum_i \frac{1}{2} m \dot{x_i}^2 \Rightarrow \frac{\partial L}{\partial \dot{x}_i} = m\dot{x_i} \text{ is conserved.} \tag{38}$$

We define the generalized momentum associated with q_i as

$$P_j = \frac{\partial L}{\partial \dot{q}_j} \tag{39}$$

Example 11 Ball in 3D

Consider the following system:



Figure 6: Ball in 3D

We have the Lagrangian as:

$$l = \frac{1}{2} m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz \Rightarrow P_x \text{ and } P_y \text{ are conserved.} \tag{40}$$

Example 12 Spherical Pendulum

Consider the following system:

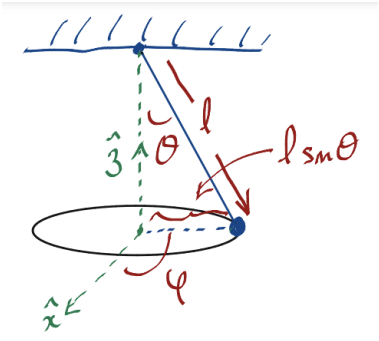


Figure 7: Spherical Pendulum

Then the Lagrangian of the system would be

$$L = \frac{1}{2}ml^2(\dot{\theta}^2 + \dot{\varphi}^2 \sin^2\theta) + mgl\cos\theta \tag{41}$$

Therefore φ is cyclic:

$$P_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = ml^2 \sin^2\theta\dot{\varphi} \text{ is conserved.} \tag{42}$$

Note: In general, generalized momentum $\neq$ linear momentum!

Noether's Theorem

To prove Noether's Theorem, we need to define a symmetry. Consider a transformation of the generalized coordinates of a system parametrized by λ:

$$q_i(t) \to Q_i(t, A) \text{where } Q_i(t, 0) = q_i(t)$$

We say transformation is a symmetry if the EOM for the Q_i's are the same as for the q_i's. We claim a transformation is a symmetry if:

$$\frac{\partial L}{\partial \lambda}(Q_i(t,\lambda), \dot{Q}_i(t,\lambda), t) = \frac{dF}{dt} \text{ for the same function } F(Q_i, \dot{Q}_i, t) \tag{43}$$

The key is to understand any change in the Lagrangian should not affect the equations of motion derived from it. Remember adding a total time derivative of a function F(dF/dt) to the Lagrangian does not change the equations of motion.
Therefore if the change of the Lagrangian under the transformation is given by $\frac{dF}{dt}$, then the transformation represents a symmetry of the system because it leaves the equations of motion invariant.

Proof.

$$\frac{\partial L(Q_i(t,\lambda), \dot{Q}_i(t,\lambda), t)}{\partial \lambda}\Big|_{\lambda=0} = \sum_i \Big(\frac{\partial L}{\partial Q_i}\frac{\partial Q_i}{\partial \lambda} + \frac{\partial L}{\partial \dot{Q}_i}\frac{\partial \dot{Q}_i}{\partial \lambda}\Big)\Big|_{\lambda=0} \tag{44}$$

$$\sum_i \Big(\frac{\partial L}{\partial q_i}\frac{\partial Q_i}{\partial \lambda} + \frac{\partial L}{\partial \dot{q}_i}\frac{\partial \dot{Q}_i}{\partial \lambda}\Big)\Big|_{\lambda=0}$$

Since $\frac{\partial L}{\partial q_i} = \frac{d}{dt}(\frac{\partial L}{\partial \dot{q}_i})$, We have $\frac{\partial \dot{Q}_i}{\partial \lambda} = \frac{d}{dt}(\frac{\partial Q_i}{\partial \lambda})$ Thus we arrive with:

$$\frac{\partial L}{\partial \lambda}\Big|_{\lambda=0} = \sum_i \frac{d}{dt}\Big(\frac{\partial L}{\partial \dot{q}_i}\frac{\partial Q_i}{\partial \lambda}\Big)\Big|_{\lambda=0} = \frac{dF}{dt}$$

$$\frac{d}{dt}\Big(\sum_i P_i \frac{\partial Q_i}{\partial \lambda}\Big|_{\lambda=0} - F\Big) = 0 \tag{45}$$

We now have our conserved quantity inside the bracket. □

Example 13 Spatial Translations

The Lagrangian of a closed system is invariant if the coordinate of all the particle are shifted by the same amount in some direction. For example, N particles interacting in a potential which only depends on the relations displacements of the particles:

$L = \frac{1}{2}\sum_i m_i \dot{\vec{r_i}}^2 - V(\vec{r_i} - \vec{r_i})$ which the latter term only depends on difference of displacements

To find the conserved quantity, we refer to equation(60), so we have:

$$\frac{\partial \vec{Q_a}}{\partial \lambda} = \hat{n}$$

$$\sum_{particle} \frac{\partial L}{\partial \dot{\vec{q_i}}} \cdot \hat{n_i} = \sum_i m\vec{v}\hat{n} = \vec{P} \cdot \hat{n} \text{ is conserved} \tag{46}$$

The total linear momentum of the system is $\vec{P} = \sum_{particle} \vec{p_i} = \sum_i m\vec{v_i}$ So space translation invariance $\Longleftrightarrow$ (linear) momentum conservation.

Example 14 Time Translation

Note time is not a generalized coordinate, so the transformation is not $t \to t + \lambda$ but rather

$$q_i(t) \to Q_i(t, \lambda) = q_i(t + \lambda)$$

In general, L depends on t in 2 ways:

1. via the generalized coordinate & velocities
2. explicitly (for example, a time dependent potential energy, like the pendulum or a rotating disk)

The conserved quantity associated with time translation symmetry is the energy of the system. this is true whether or not the Lagrangian explicitly depends on time. We will begin by showing the proof when Lagrangian has no explicit dependency on time.

Proof. We start by considering how the Lagrangian L' under this transformation is given by:

$$L'(Q_i(t, \lambda), \dot{Q_i}(t, \lambda), t) = L(q_i(t + \lambda), \dot{q_i}(t + \lambda), t + \lambda)$$

To determine the change of the Lagrangian under this transformation, we expand the right-hand side to first order in λ using Taylor series expansion around $\lambda = 0$:

$$L'(Q_i(t, \lambda), \dot{Q_i}(t, \lambda), t) \simeq L'(q_i(t, \lambda), \dot{q_i}(t, \lambda), t)$$

where

$$\frac{dL}{dt} = \frac{\partial L}{\partial t} + \frac{\partial L}{\partial q_i}\frac{\partial q_i}{\partial t} + \frac{\partial L}{\partial \dot{q_i}}\frac{\partial \dot{q_i}}{\partial t}$$

We can simplify this by noting that $\frac{dq_i}{dt} = \dot{q_i}$ and that $\frac{\partial L}{\partial \dot{q_i}}\frac{d\dot{q_i}}{dt} = \frac{d(\partial L/\partial \dot{q_i} \cdot \dot{q_i})}{dt} - \dot{q_i} \cdot \frac{d(\partial L/\partial \dot{q_i})}{dt}$ Substituting these and using Euler-Lagrange equation, we find that the Lagrangian changes by a total time derivative:

$$\frac{dL}{dt} = \frac{\partial L}{\partial t} + \frac{d}{dt}(\frac{\partial L}{\partial \dot{q_i}} \cdot \dot{q_i}) - \frac{dL}{dt}$$

We can simplify this to get:

$$\frac{dL}{dt} = \frac{dH}{dt}$$

where

$$H = \frac{\partial L}{\partial \dot{q_i}} \cdot \dot{q_i} - L \tag{47}$$

is the Hamiltonian, which for many physical systems represents the total energy of the system. The conserved quantity for time translation symmetry, therefore, is the Hamiltonian or the total energy of the system. □

Now, if the Lagrangian has an explicit time dependence, then it means the system isn't symmetric under time translation. In other words, the physics of the system changes as time progresses.
For instance, consider a charged particle in an oscillating electric field. The electric field (and hence the potential energy of the particle) explicitly depends on time. If you were to "translate" the system forward in time (i.e., observe it at a later time), the system's dynamics would not look the same, because the field has changed.
This has implications for the conserved quantities. Noether's theorem links symmetries of a system with conserved quantities, but when the symmetry is broken (as is the case with time-dependent Lagrangians), the associated quantity is no longer conserved.
With explicit time dependence in the Lagrangian, the total energy of the system (as given by the Hamiltonian) isn't conserved. This makes sense intuitively: if you have an external, time-varying force acting on your system, it can do work on the system, thereby changing the system's total energy.
However, even when the Lagrangian has explicit time dependence, Noether's theorem gives us a related quantity that is still conserved. This quantity is the "generalized energy", defined as:

$$E = \frac{\partial L}{\partial \dot{q}_i} \cdot \dot{q}_i - L + \frac{\partial L}{\partial t} \tag{48}$$

which is conserved when the Lagrangian has explicit time dependence, i.e. when $\frac{\partial L}{\partial t} \neq 0$.

Euler's Theorem on Homogeneous Functions

If $f(\alpha x_1, \alpha x_2, ...\alpha x_N) = \alpha^n f(x_1, x_2, ...x_N)$ for some n, then $\sum_i x_i \frac{\partial f}{\partial x_i} = nf(x)$. In the case of Noether's Theorem, we have:

$$\sum_i \dot{q}_i \frac{\partial T}{\partial \dot{q}_i} = 2T \text{ if T is quadratic in } \dot{q}_i \tag{49}$$

Therefore we can show

$$H = \sum_i \dot{q}_i \frac{\partial T}{\partial \dot{q}_i} - L = 2T - L = 2T - T + V = T + V \text{ which is the total energy} \tag{50}$$

Therefore we know as long as T is quadratic in generalized velocities, $H = T + V = E$.

Example 15 Rotationally Invariant System

Consider an infinitesimal rotation by an angle $\delta\lambda$ about an axis $\hat{n}$:

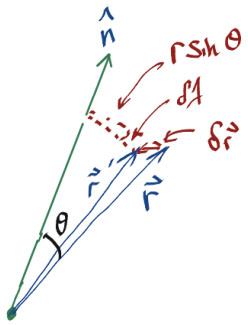


Figure 8: Rotationally Invariant System

We have $\delta\vec{r} \perp \vec{r}$ & $\delta\vec{r} \perp \hat{n}$ and $\delta\lambda = \frac{|\delta\vec{r}|}{r \sin\theta}$ Thus we arrive with

$$\delta\vec{r} = \hat{n} \times \hat{r}\ \delta\lambda$$

so $\frac{d\vec{r}}{d\lambda} = \hat{n} \times \hat{r}$ and $F = 0$ (L unchanged). Therefore we have

$$\frac{d}{dt} \sum_i \vec{P}_i \cdot \frac{d\vec{r}}{d\lambda}\Big|_{\lambda=0} = \frac{d}{dt} \sum_i \vec{P}_i \cdot (\hat{n} \times \hat{r_i}) = 0 \tag{51}$$

A useful vector multiplication property is

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{c} \times \vec{a}) \tag{52}$$

Using this property we eventually arrive with

$$\sum_i \vec{P}_i \cdot (\hat{n} \times \hat{r}) = \hat{n} \cdot \sum_i \vec{r_i} \times P_i = \hat{n} \cdot \vec{M} \text{ is conserved} \tag{53}$$

where $\hat{M} \equiv \sum_i \vec{r_i} \times \vec{P_i}$ is the angular momentum of the system. Meaning if a system is invariant under rotations about the $\hat{n}$ axis, the $\hat{n}$ component of $\hat{M}$ is conserved.

Comments

1. Noether's Theorem is applicable to any conserved quantity.
2. If a system is not closed, L may depend on some coordinate but not others $\Rightarrow$ some components of $\vec{P}$ & $\vec{M}$ may be conserved while others aren't.
3. Consider a ball falling down a 3-dimensional space. The system of the ball alone is not closed, so P_z of ball is not conserved (P_x & P_y are). Note if we ignored everything else in the universe, Earth + ball is a closed system ($P_z^{ball} + P_z^{Earth}$ is conserved).
4. Newton's 3rd law is just conservation of momentum. For every action, there is an equal and opposite reaction. For a translational symmetry, where the transformation is a shift in position: $r_i \to r_i + \lambda\hat{n}$, the corresponding conserved quantity according to Noether's Theorem is linear momentum. If the transformation leaves the Lagrangian invariant (which it does for spatial translations, i.e., moving the whole system without changing its internal state), then the derivative of the Lagrangian with respect to λ must be zero. For Cartesian coordinates, the Lagrangian's dependence on position comes only through the potential energy term, since the kinetic energy term depends only on the velocities, not on the positions directly ($L = T - V$).

   $$\frac{\partial L}{\partial \lambda} = \sum_i \frac{\partial L}{\partial r_i} = -\sum_i \frac{\partial V}{\partial r_i} = \sum_i F_i = 0 \tag{54}$$

   Therefore this satisfies Noether's Theorem such that $\frac{\partial L}{\partial \lambda} = 0$.
5. $\vec{P}$ & $\vec{M}$ are additive such that $\vec{M}total = \sum_i \vec{M}_i$ and $\vec{P}_{total} = \sum_i \vec{P}_i$. However, E is only additive for free particles (in general $e \neq \sum_i E_i$ because of interactions.)
6. P, $\vec{M}$, & E clearly depend on the reference frame (we will transform between frames when needed), but $\vec{M}$ also depends on the choice of origin:

   $$\vec{r} \to \vec{r} + \vec{a}$$
   $$\vec{M}_i = \vec{r_i} \times \vec{p_i} \to (\vec{r_i} + \vec{a}) \times \vec{p_i} = \vec{M}_i + \vec{a} \times \vec{p_i}$$
   $$\therefore \vec{M} \to \vec{M} + \vec{a} \times \vec{P} \text{ if origin is shifted by } \vec{a}$$
7. We can relate the angular momentum of an object about an arbitrary point to the angular momentum about the COM by $\vec{M} = \vec{M}_{com} + \vec{R}_{com} \times \vec{P}$

Integrals of Motion

Consider a free particle in 3D: $l = \frac{1}{2}m\vec{v}^2$, we have E, $\vec{P}$, & $\vec{M}$ conserved $\rightarrow$ 7 quantities. Not all quantities are independent, which is why the initial state of the system is completely determined by 6 quantities:

$$\vec{r}(t_0)$$
$$\vec{v}(t_0)$$

For instance, we have $E = \frac{\vec{P}^2}{2M}$ and $\vec{P} \cdot \vec{M} = 0$ (Since $\vec{M} = \vec{r} \times \vec{P} \perp \vec{P}$), so there are only 5 independent conserved quantities. In fact, this is the maximum number of conserved quantities that a single particle can have. More generally: A system with d degrees of freedom can have at most 2d-1 independent integrals of motion (conserved quantities). Given a system with d degrees of freedom, the state of the system at any given time can be fully specified by 2d variables: d generalized coordinates and d generalized velocities. These 2d variables can be regarded as the coordinates of a point in a 2d-dimensional space, known as the phase space of the system.
An integral of motion (conserved quantity) in classical mechanics is a function of the coordinates and velocities that remains constant as the system evolves in time. From a geometrical perspective, a conserved quantity defines a hypersurface in the phase space: all the points on this hypersurface have the same constant value of the conserved quantity.
As the system evolves, the point representing the system state moves in phase space, but if a quantity is conserved, it remains on the hypersurface defined by that conserved quantity. If we have multiple conserved quantities, the motion of the system is constrained to the intersection of the corresponding hypersurfaces.
Let's now consider why there can be at most 2d-1 independent integrals of motion.
The reason for this limit is that the equations of motion must allow at least one degree of freedom for the system to evolve in time. Remember that we're considering a system with 2d degrees of freedom in phase space, so it can a priori move in any of these directions. Each independent conserved quantity constrains the motion to a subspace one dimension smaller. However, if we had 2d independent conserved quantities, we would constrain the system state to a zero-dimensional subspace, i.e., a single point. This would mean that the system can't evolve, which contradicts the dynamical nature of physical systems.
Therefore, we can have at most 2d-1 independent conserved quantities, leaving at least one degree of freedom for the system to evolve. The "-1" essentially accounts for the time evolution of the system.

Example 16 1D Motion

We have

$$\frac{1}{2}m\dot{x}^2 - u(x)$$

Thus the E-L equation is:

$$m\ddot{x} = -\frac{du(x)}{dx} \text{ which is very difficult to solve.}$$

Instead, we can use conservation laws and integral of motion: $H = E = \frac{1}{2}m\dot{x}^2 + u(x)$, this gives a trajectory in $x, \dot{x}$ space. Using this conserved quantity, we can analyze the motion of the particle in terms of its position x and velocity $\dot{x}$ instead of solving a differential equation:

$$\dot{x} = \frac{dx}{dt} = \sqrt{\frac{2(E-u(x))}{m}} \Rightarrow dt = \sqrt{\frac{m}{2}}\frac{dx}{\sqrt{E-u(x)}}$$
$$t = \sqrt{\frac{m}{2}}\int \frac{dx}{\sqrt{E-u(x)}} + C$$

Let's get some intuition for its behaviours (which will be useful going forward).

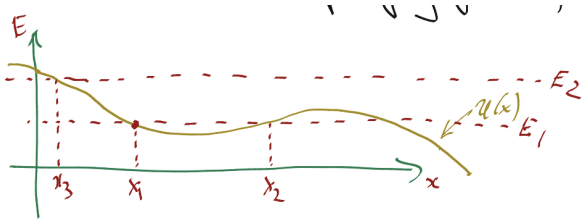


Figure 9: u(x)

Suppose $u(x)$ is given by the curve in the figure, if $E = E_1$, the particle will oscillate back & forth between x_1 & x_2 with period $T = \sqrt{2m}\int_{x_1(E)}^{x_2(E)} \frac{dx}{\sqrt{E-u(x)}}$, where x_1 & x_2 are called turning points. At the turning points, $T = 0 \Rightarrow E = u(x_1) = u(x_2)$. if $E = E_2$, the particle will go from x_3 & escape to ∞.

Example 17 Simple Pendulum

Consider the following system:

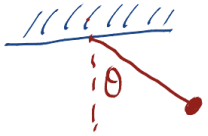


Figure 10: Simple Pendulum

We have

$$E = \frac{1}{2}ml^2\dot{\theta}^2 - mgl\cos\theta \tag{55}$$

Let's solve for arbitrary θ: Start from rest at $\theta = \theta_0$ where

$$E = -mgl\cos\theta_0$$
$$\frac{1}{2}ml^2\dot{\theta}^2 - mgl\cos\theta = -mgl\cos\theta = -mgl\cos\theta_0$$
$$\Rightarrow \dot{\theta} = \sqrt{\frac{2g}{l}(\cos\theta - \cos\theta_0)}$$

So we can write an expression for the period, which is

$$\frac{d\theta}{dt} = \sqrt{\frac{2g}{l}(\cos\theta - \cos\theta_0)} \Rightarrow T = 4\sqrt{\frac{l}{2g}}\int_0^{\theta_0} \frac{d\theta}{\cos\theta - \cos\theta_0}$$

Let's show that in the limit of small oscillations we get the familiar result:

$$\theta,\ \theta_0 << 1 \Rightarrow \cos\theta = 1 - \frac{\theta^2}{2!} + o(\theta_4)$$
$$\therefore (\cos\theta - \cos\theta_0)^{-1/2} = (\frac{\theta^2}{2} - \frac{\theta_0^2}{2})^{-1/2} + ... = \frac{\sqrt{2}}{\theta_0}(1 - \frac{\theta^2}{\theta_0^2})^{-1/2} + ...$$

Defining $x \equiv \frac{\theta}{\theta_0}$,

$$T = 4\sqrt{\frac{l}{2g}} \times \sqrt{2}\int_0^1 \frac{1}{\sqrt{1-x^2}}dx + ... = 2\pi\sqrt{\frac{g}{l}} + ... \text{ as expected} \tag{56}$$

Exercise: Carry out the Taylor Expansion to the next order, to get $T = 2\pi\sqrt{\frac{l}{g}}(1 + \frac{1}{16}\theta_0^2 + ...)$

Example 18 Two Particles in a Central Potential

We have $u(\vec{r_1}, \vec{r_2}) = u(|\vec{r_1}, \vec{r_2}|)$, and the corresponding Lagrangian is:

$$L = \frac{1}{2}m_1\vec{v_1}^2 + \frac{1}{2}m_2\vec{v_2}^2 - u(|\vec{r_1}, \vec{r_2}|)$$

Although at first glance this seems to have 6 d.o.f, this is a closed system, meaning the COM moves with a constant velocity, therefore we can choose an inertial frame and choose $\vec{R}_{com}$ at the origin. Define $\vec{r} \equiv \vec{r_1} - \vec{r_2}$ and since $m_1\vec{r_1} + m_2\vec{r_2} = m_1\vec{r} + (m_1 + m_2)\vec{r_2} = 0$, we get

$$\vec{r_2} = \frac{-m_1}{m_1 + m_2}\vec{r} \;\; \& \;\; \vec{r_1} = \frac{m_2}{m_1 + m_2}\vec{r} \tag{57}$$

$\therefore$ In the COM frame, we can express both positions in terms of the relative separation $\vec{r}$

$$L = \frac{1}{2}\dot{\vec{r}}^2(\frac{m_1m_2}{m_1 + m_2}) - u(r) \text{ where } r \equiv |\vec{r}| \tag{58}$$

This is equivalent to a single particle of mass with reduced mass $\mu = \frac{m_1m_2}{m_1+m_2}$ in an external potential u(r). Note the system is not closed since the potential depends on distance from the origin. So rather than studying the 2-body problem, we can study the equivalent problem of 1 particle of reduced mass μ in a central external potential.

$$L = \frac{1}{2}\mu\dot{\vec{r}}^2 - u(r) \tag{59}$$

So we reduced the system to 3 d.o.f $\Rightarrow$ maximum 5 independent integrals of motion. Which terms are conserved in this case?

1. $\vec{P}$ is not invariant under translation in any direction since the origin in fixed.
2. $\vec{M}$ is invariant under rotations about any axis through the origin (so only $\vec{M}$ about the origin is conserved).
3. H is invariant since kinetic energy T is quadratic in $\dot{\vec{r}}$ $\therefore H = E$

Since $4 > 3$, the system in integrable and we can use conservation laws to reduce it to quadratures (solutions in integral form) without solving the equations of motion. Let's see how to do this: $\vec{M} = \vec{r} \times \vec{P}$ is constant $\Rightarrow \vec{r}$ is always $\perp$ fixed vector $\vec{M} \Rightarrow \vec{r}$ moves in a plane $\perp \vec{M}$.

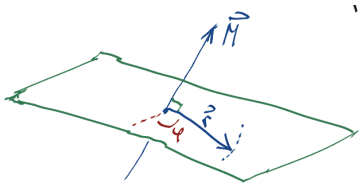


Figure 11: Perpendicular Relationship between $\vec{M}$ and $\vec{r}$

Choose coordinates such that $\vec{M} = M_z\vec{z}$, then the motion is always in the xy plane $\Rightarrow$ in spherical polar coordinates, $\theta = \pi/2$ is fixed, and we are left with 2 d.o.f, that being r and φ

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2) - u(r) \tag{60}$$

So $\vec{M} = M_z\hat{z}$ is conserved, $M_z = (\vec{r} \times \vec{P})_z = mr^2\dot{\varphi} = constant$ $\therefore \dot{\varphi} = \frac{M_z}{mr^2} \propto \frac{1}{r^2}$. This means in a time dt, particle moves an angle $d\varphi$, sweeping out an area $dA = \frac{1}{2}r^2d\varphi$

$$dA = \frac{1}{2}r^2d\varphi = \frac{1}{2}r^2\frac{d\varphi}{dt} \Rightarrow \frac{dA}{dt} = \frac{1}{2}r^2\dot{\varphi} \tag{61}$$

Since $M_z = mr^2\dot{\varphi}$ is constant, we can represent $\dot{\varphi}$ as $\dot{\varphi} = \frac{M_z}{mr^2}$ Therefore $\Rightarrow \frac{dA}{dt} = \frac{1}{2}r^2 \times \frac{M_z}{mr^2} = \frac{M_z}{2m}$ is constant! This is Kepler's 2nd law, which holds for any central potential. The third is only true for a $1/r$ potential, as is the first (as we shall see shortly). We can now use conservation of angular momentum to write the Hamiltonian for this system in a very slick way:

$$H = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2) + u(r) \tag{62}$$

$$= \frac{1}{2}m\dot{r}^2 + \frac{M_z^2}{2mr^2} + u(r) \tag{63}$$

Now, if we define $U_{eff}(r) \equiv u(r) + \frac{M_z^2}{2mr^2}$, we arrive with the final result

$$\therefore H = \frac{1}{2}m\dot{r}^2 + u_{eff}(r) \tag{64}$$

Why is this so slick? Writing it this way gets rid of $\dot{\varphi}$ entirely, and H looks like the Hamiltonian for a one-dimensional problem with potential $u_{eff}(r)$. Since we know how to solve 1d motion problems, we can solve for motion in r!

Example 19 Attractive Potential

Consider the following curve

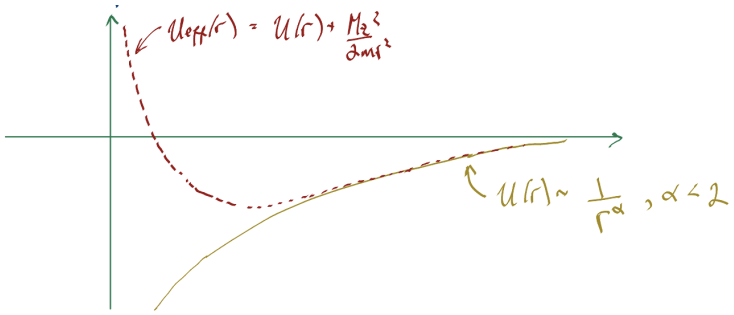


Figure 12: Attractive Potential

The centrifugal energy $\frac{M_z^2}{2mr^2}$ dominates at $r \to 0$ but is negligible at long distance. If the total energy $E < 0$, the particle will oscillate back & forth between the turning points, called a bound orbit:

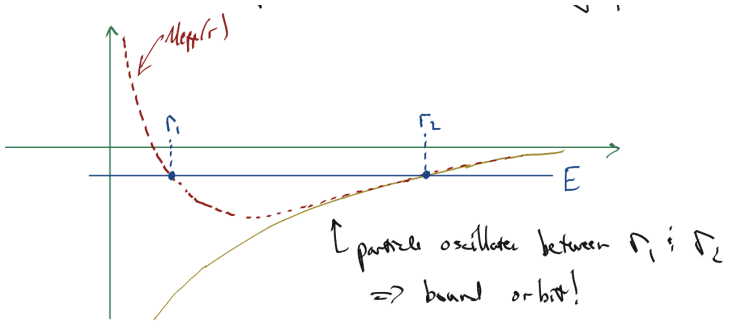


Figure 13: Bound Orbit

So we have

$$\frac{dr}{dt} = \sqrt{\frac{2}{m}(E - u_{eff}(r))} \Rightarrow t = \int \frac{1}{\sqrt{\frac{2}{m}(E - u_{eff}(r))}} dr \tag{65}$$

In 3D, particle moves in a plane and $\dot{\varphi}$ evolves with r such that $M_z = mr^2\dot{\varphi}$ stays constant $\Rightarrow$ completely determines motion. Note that in general orbits are not closed because orbits are only closed if φ goes from θ to 2π (or multiples of 2π) in the time it takes r to oscillate between r_1 & r_2 (and for a general potential, there's no reason for those 2 periods to be the same). To find the shape of the orbits, we need to solve for φ as a function of r: $\varphi(r)$

$$\frac{dr}{dt} = \frac{dr}{d\varphi}\frac{d\varphi}{dt} = \frac{M_z}{mr^2}\frac{dr}{d\varphi}$$

$$\varphi(r) = \int \frac{M_z}{r^2}\frac{dr}{\sqrt{2m(E-u_{eff}(r))}} = \int \frac{M_z}{r^2}\frac{dr}{\sqrt{2m(E-u(r)) - \frac{M_z^2}{r^2}}}$$

which can be solved if we know u(r). The change in φ in one period is

$$\Delta\varphi = 2\int_{r_{min}}^{r_{max}} \frac{M_z}{r^2}\frac{1}{\sqrt{2m(E-u(r)) - \frac{M_z^2}{r^2}}} \tag{66}$$

Note: If a particle's energy is such that it is at the minimum of $u_{eff}(r)$, its motion is simple $\Rightarrow$ circular orbit (because is is stationary in r).

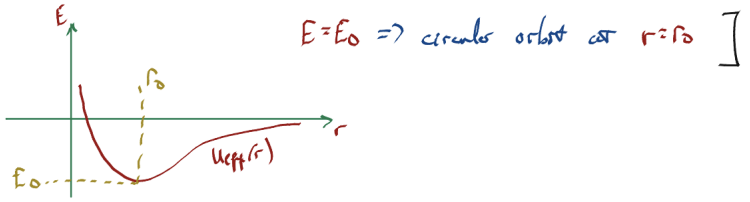


Figure 14: Circular Orbit

Notes:

1. The $\frac{M_z^2}{2mr^2}$ term in $u_{eff}(r)$ is "really" kinetic, not potential energy! It just effects the fact that as the partcile moves nearer the origin, it must rotate faster ($\dot{\varphi}$ grows) to keep M_z constant. H only cares about the total energy, so it's equivaent to a 1D problem in r where $\frac{M_z^2}{2mr^2}$ is potential energy;
2. Recall that L does distinguish T & V, so we must be careful not to substitute $\dot{\varphi}$ with $\frac{M_z}{mr^2}$ into L to get the EOM!
3. Since the centrifugal term $\frac{1}{r^2}$ as $r \to 0$, in general two particle in a central potential will never spiral in and collide $\Rightarrow$ their relative angular momentum won't allow it! Unless the potential $u(r)$ $\frac{-1}{r^{2+n}}$, $n > 0$ as $r \to 0$, in which case $u_{eff} \to -\infty$ as $r \to 0$.
4. For the case $u(r) = 0 (\& M_z \neq 0)$, we have

$$u_{eff}(r) = \frac{M_z^2}{2mr^2} \to \text{ purely repulsive (effective) potential.} \tag{67}$$

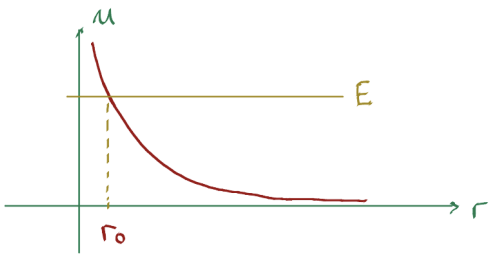


Figure 15: u(r)=0

5. Any particle with finite energy E initially moving toward the origin will turn around at a distance r_0 and then move away. This is just a particle moving in a straight line in 2D, expressed in polar coordinate!

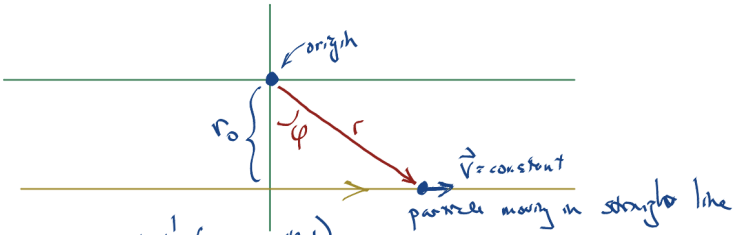


Figure 16: Particle moving Straight Line in 2D

You can check that $M_z = |\vec{r} \times \vec{P}| = r_0 m v \Rightarrow r_0 = \frac{M_z}{mv}$ but $E = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2E/m}$ $\therefore r_0 = \frac{M_z}{\sqrt{2mE}}$

Example 20 Special Case: $\frac{1}{r}$ Potential

Consider the following plot

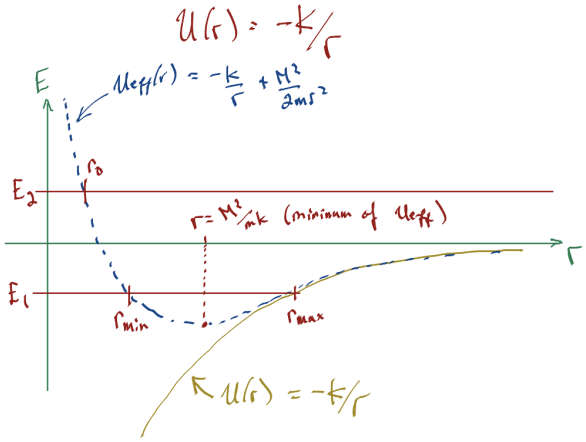


Figure 17: 1/r Potential

1. $E = E_1$ (< 0): particle oscillates between $r = r_{min}$ & $r = r_{max}$ as it orbits
2. $E = E_2$ (> 0): Unbound orbit $\rightarrow$ particle approaches origin, but at $r = r_0$ it moves off to ∞
   We can solve for the shape of the orbits directly from conservation of energy/angular momentum (reduced to quadratures), but we can use a new conservation law to do this algebraically.
   "Laplace-Runge-Lenz" vector

$$\vec{A} = \vec{P} \times \vec{M} - mk\hat{r} \text{ is conserved} \tag{68}$$

This is related to a hidden symmetry of the problem- rotations on a 4D hyperplane (!)
We can use

$$\frac{d\vec{A}}{dt} = 0 \tag{69}$$

(Note: $\vec{A} \cdot \vec{M} = 0$ & $|\vec{A}|^2$ is determined by E & $\vec{M}^2$. So this only adds 3-2=1 new integral of motion).
This new conservation law will allow us to solve the system algebraically $\rightarrow$ no integrals required!

What is $\vec{A}$?

First consider the point of the nearest approach ($r = r_{min}$): at this point $\vec{P} \perp \vec{r}$ (since otherwise it could be moving closer/further):

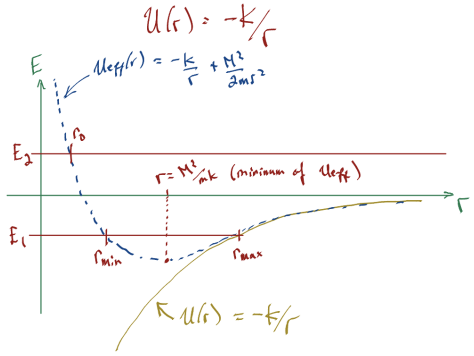


Figure 18: Direction of $\vec{A}$

$\therefore \vec{A}$ points radially out, as shown in the figure and since $\vec{A}$ is conserved, it must be the same everywhere in the orbit. For example, at $r = r_{max}$:

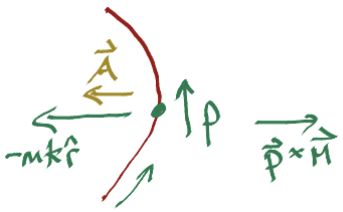


Figure 19: $r = r_{max}$

Now $\vec{P}$ & $\vec{P} \times M$ point in the opposite direction, as does $-mk\hat{r}$ but their sum is unchanged. Since $mk\hat{r}$ has the same negligible (and $\vec{M}$ is unchanged), $|\vec{P}|$ must be smaller $\rightarrow$ moving slower. Since the location of each point in the orbit gives you $-mk\hat{r}$ and the direction of $\vec{P}$, at each point the magnitude of $\vec{P}$ must be such that $\vec{A} = \vec{P} \times \vec{M} - mk\hat{r}$ is conserved:

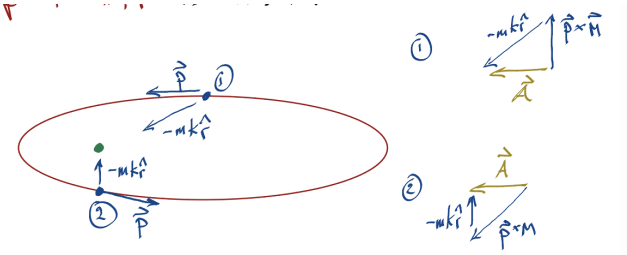


Figure 20: Analysis on $\vec{A}$

At r_{min} & r_{max}, $\vec{A}$ is perpendicular to path (since $\vec{P}$ is along the path, $\hat{r} \perp$ path). Therefore orbit is closed. On the shape of the orbit, we can define φ to be the angle between $\vec{A}$ & $\vec{r}$, then by definition, we have

$$\vec{A} \cdot \vec{r} = AR\cos\varphi = \vec{r} \cdot (\vec{P} \times \vec{M}) - mkr = M^2 - mkr = M^2$$

$$\therefore r(A\cos\varphi + mk) = M^2$$

$$\frac{1}{r} = \frac{mk}{M^2}(1 + e\cos\varphi) \text{ where } e \equiv \frac{A}{mk} = \sqrt{1 + \frac{2EM^2}{mk^2}}$$

This is the equation for a conic section (ellipse, parabola or hyperbola) of eccentricity e!

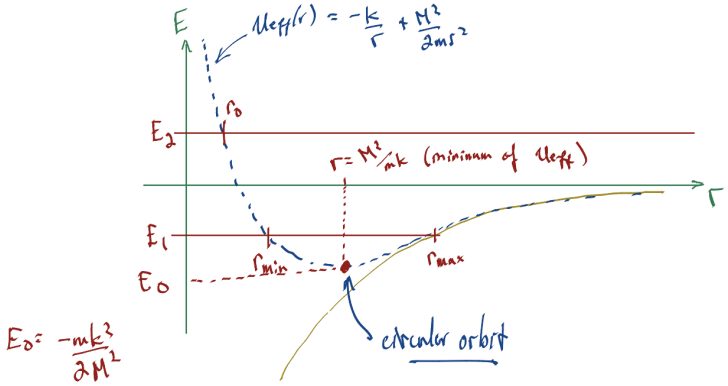


Figure 21: E vs. r

So an object in a circular orbit has the lower possible energy consistent with its angular momentum. As we increase E, the particle will oscillate between r_{min} & r_{max} → elliptical orbit of eccentricity.

$$e = \sqrt{1 + \frac{2EM^2}{mk^2}} = \sqrt{1 - \frac{E}{E_0}} \tag{70}$$

When we hit $E = 0$, the orbit is no longer bound (e=1 (parabola) and $\frac{1}{r} \to 0$ at $\varphi = \pi \ \ \therefore r \to \infty$). When $E > 0$, $r \to \infty$ for $\varphi \equiv \varphi_0 < \pi \Rightarrow$ hyperbolic orbit.

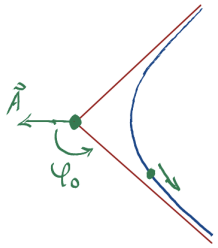


Figure 22: Hyperbolic Orbit

One final comment: since $e\frac{A}{mk}$, we could equally well have defined the conserved "eccentricity vector"

$$\vec{e} \equiv \frac{\vec{A}}{mk} \tag{71}$$

which has a transparent physical interpretation: it is a conserved vector whose magnitude is the eccentricity of the orbit, and which points in the outward radial direction at the point of closest approach ("periapsis").

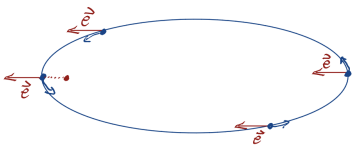


Figure 23: Periapsis

2 Dynamics of Rigid Bodies

Example 1 To set the Stage

In first year, you looked at rotations about a fixed (usually symmetric) axis, such as

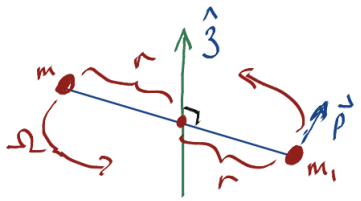


Figure 24: Two masses on a rod, Rotating about an axis perpendicular to the rod

In this case, we write $M_z = I_z\Omega$ For both masses $\vec{r} \times \vec{P}$ is in the $\hat{z}$ direction, and $|\vec{r} \times \vec{P}| = rp = rmv_\theta = rmr\Omega = mr^2\Omega$ $\therefore$ the total angular momentum (assume massless rod) is

$$\vec{M} = M_z\hat{z} \ , M_z = 2mr^2\Omega \equiv I_z\Omega \tag{72}$$

Let's take this same setup, but instead rotate the rod about a less symmetrical axis:

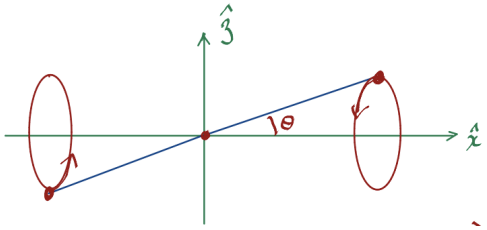


Figure 25: Rotate rod about $\hat{x}$ axis (θ fixed)

To specify the axis of rotation, define $\vec{\Omega}$ at a vector with magnitude Ω, direction along axis of rotation with direction of motion given by the right-hand rule...so here, $\vec{Omega} = \Omega\hat{x}$. What about $\vec{M}$? Consider the mass on the right...then we have (in the position shown). Since $\vec{M}$ for the left mass is identical, we arrive with

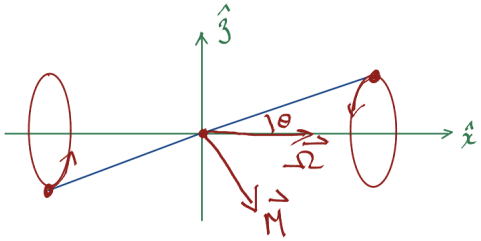


Figure 26: Axis of Rotation for the 2 Masses

So in this case, $\vec{M}$ & $\vec{\Omega}$ are not parallel! Therefore the relationship between $\vec{M}$ & $\vec{\Omega}$ is more complicated then for rotations about the $\hat{z}$ axis. Also note that $\vec{M}$ rotates in a circle as the rod rotates. Therefore $\vec{M}$ is not constant.

What is Rigid Body?

By definition, a rigid body system is an arrangement of masses (discrete or continuous) which do not move relative to each other.

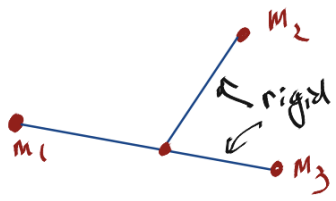


Figure 27: Example of Rigid Body System

A rigid body has 6 degrees of freedom. To set this up, it's convenient to consider 2 natural frame of reference: an inertial frame (lab frame) and a moving reference frame (body-fixed frame) in which the rigid body is at rest. We will denote X,Y,Z as coordinates in lab frame, and x,y,z as coordinates in the body-fixed frame. Note any point on the body has a fixed position in the x,y,x coordinates. It requires 3 angles to specify the orientation of x,y,x with respect to X,Y,Z. Now we will work out kinematics in these variable. Consider an arbitrary point P on the body and call the displacement of P in the lab/fixed frame $\vec{P}$, then we can write $\vec{P}$ as the same as the displacement of a fixed point O on the body, and the displacement of P w.r.t O is $\vec{P} = \vec{R} + \vec{r}$.

Now, in an infinitesimal time dt, the point P will move by a displacement $d\vec{P}$. Since $|\vec{r}|$ is fixed, this motion can be split into

1. displacement $d\vec{R}$ of O (motion of the object as a whole)
2. rotation $d\varphi$ about an axis $\hat{n}$ passing through O $()d\vec{\varphi} \equiv \hat{n}d\varphi$ is a vector pointing along the axis of rotation, with direction of rotation given by right hand rule).

Then the total displacement of P can be written as

$$d\vec{P} = d\vec{R} + d\vec{r} \tag{73}$$

$$= d\vec{R} + d\vec{\varphi} \times \vec{r} \tag{74}$$

Therefore $\frac{d\vec{P}}{dt} = \frac{d\vec{R}}{dt} + \frac{d\vec{\varphi}}{dt} \times \vec{r}$

$\therefore \vec{v} = \vec{V} + \vec{\Omega} \times \vec{r}$ So if O is the COM, $\vec{V}$ is the velocity of the COM, and Ω is the angular velocity of the object about the COM.
Note this formula is true for any fixed point O on the body.

Example 2 Consider a New Point

Since this is true for any point O, consider a new point o', a displacement $\vec{a}$, then we have:

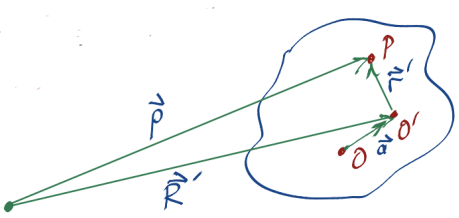


Figure 28: New point O'

Then $\vec{P} = \vec{R} + \vec{r} = \vec{R}' + \vec{r}' = \vec{R} + \vec{a} \times \vec{r}'$
So $\vec{v} = \vec{V} + \vec{\Omega} \times (\vec{r}' + \vec{a}) = \vec{V} + \vec{\Omega} \times \vec{a} + \vec{\Omega} \times \vec{r}'$
But we also know that $\vec{P} = \vec{R}' + \vec{r}' \Rightarrow \vec{v} = \vec{V}' + \vec{\Omega}' \times \vec{r}'$
And since this is true for arbitrary $\vec{a}$, we must have

$$\vec{V}' = \vec{V} + \vec{\Omega} \times \vec{a} \text{ and } \vec{\Omega}' = \vec{\Omega} \tag{75}$$

So if we take O to be the COM of the object, we know

1. The instantaneous angular velocity of the body (or of a frame fixed to the body) is independent of its origin.
2. We can choose a point o' for which $\vec{v}' = 0 \Rightarrow$ motion of the body is pure rotation about an axis through O' $\Rightarrow$ "instantaneous axis of rotation".

Example 3 Rolling Wheel (without slipping)

Consider the following system

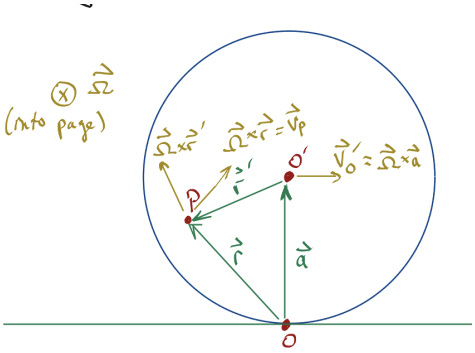


Figure 29: Rolling Wheel

Note: Since O is in contact with the ground, it's instantaneous velocity $\vec{V_0} = o$! (Otherwise it would be moving w.r.t ground $\therefore$ slipping). The math are shown here:

$$\vec{v}_p = \vec{v}_o + \vec{\Omega} \times \vec{r}$$
$$\vec{v}_{o'} + \vec{\Omega} \times \vec{r}'$$
$$\vec{v}_{o'} = \vec{v}_o + \vec{\Omega} \times \vec{a}$$
$$|\vec{a}| = R$$

$\therefore$ motion is pure rotation (instantaneously) about O

$$\vec{v}_p = \vec{v}_o + \vec{\Omega} \times \vec{r} = \vec{\Omega} \times \vec{r}$$
(but a time dt later, O is no longer at rest)
but we also have $\vec{v}_p = \vec{v}_{o'} + \Omega \times \vec{r}'$
$$= \vec{\Omega} \times \vec{a}$$

and geometrically, we have

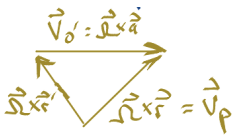


Figure 30: Geometrical Interpretation

Dynamics

We have L=T-V, let's look at the simplest situation when V=0 (no potential extended force). Then, if the body is made up of a number of discrete masses m_a, we have

$$T = \frac{1}{2}\sum_a m_a v_a^2 \tag{76}$$

We can easily generalize this to a continuous mass distribution later. If the ath particle is at the position $\vec{r_a}$ in the rotating (body-fixed) frame, then

$$\vec{v_a} = \vec{v_{sum}} + \vec{\Omega} \times \vec{r_a} \tag{77}$$

where $\vec{v}$ is the velocity of O. We can therefore write

$$\begin{aligned} T &= \sum_a \frac{1}{2} m_a \vec{v_a}^2 \\ &= \sum_a \frac{1}{2} m_a (v_s \vec{u} m + \vec{\Omega} \times \vec{r_a})^2 \\ &= \sum_a \frac{1}{2} m_a (v_s \vec{u} m^2 + 2\vec{v} \cdot (\vec{\Omega} \times \vec{r_a}) + (\vec{\Omega} \times \vec{r_a}^2)) \end{aligned} \tag{78}$$

It would be nice if we could get rid of the cross term... we can do this with 2 possible choices for O:

1. If 0 is an instantaneous axis of rotation, then $\vec{v} = 0 \Rightarrow \vec{v} \cdot (\vec{\Omega} \times \vec{r_a}) = 0$
2. If 0 is the center of mass, then

$$\begin{aligned} \sum_a m_a \vec{r_a} \cdot (\vec{v} \times \vec{\Omega}) &= (\sum_a m_a \vec{r_a}) \cdot (\vec{v} \times \vec{\Omega}) \\ &= 0 \text{ by definition of COM!} \end{aligned}$$

So we can always choose O to be one of these (usually COM) & can therefore neglect the cross term. Meanwhile, the 3rd term is

$$\frac{1}{2}\sum_a m_a (\vec{\Omega} \times \vec{r_a} \cdot (\vec{\Omega} \times \vec{r_a}))$$

Using arithmetic rule, we arrive with

$$(\vec{\Omega} \times \vec{r_a})^2 = \Omega^2 r_a^2 - (\Omega \cdot \vec{r_a})^2$$

So we get

$$T = \frac{1}{2} m v^2 + \frac{1}{2}\sum_a m_a (\Omega^2 r_a^2 - (\vec{\Omega} \cdot \vec{r_a})^2) \tag{79}$$

So we split the kinetic energy of rigid body into KE of translation & KE of rotation.

KE of Rotation

The KE of rotation looks a bit nasty. But if we play with it, we can write it in a very nice form.

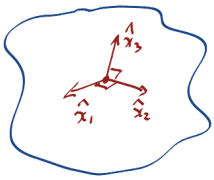


Figure 31: Rotation Axis

Define $\hat{x_i}$,$i = 1...3$ to be the unit vector in the body-fixed frame (we will use x,y,z & $x_1 = x$ $x_2 = y$ $x_3 = z$ interchangeably).

$$r_a^2 = \sum_i r_{a_i}^2 \text{ index i represents 1,2,or 3}$$

$$\begin{aligned}(\vec{\Omega} \cdot \vec{r_a})^2 &= \sum_i (\Omega_i r_{a_i})^2 \\ &= \sum_i (\Omega_i r_{a_i})(\sum_j \Omega_j r_{a_j}) \\ &= \sum_{ij} \Omega_i \Omega_j r_{a_i} r_{a_j}\end{aligned}$$

If we also write $\Omega^2 = \sum_i \Omega_i^2 = \sum_{ij} \Omega_i \Omega_j \delta_{ij}$ We can write the rotational KE as

$$\frac{1}{2} \sum_{ij} \Omega_i \Omega_j (\sum_a m_a (\delta_{ij} \vec{r_a}^2 - r_{ai} r_{aj})) \tag{80}$$

We have defined the moment of inertia of the object of the object as the set of 9 numbers inside the braket:

$$I_{ij} \equiv \sum_a m_a (\delta_{ij} \vec{r_a}^2 - r_{ai} r_{aj}) \tag{81}$$

Note that I_{ij} is a geometric characteristic of the object...it depends only on the ditribution of mass in the body (it is defined in the body-fixed frame $\rightarrow$ doesn't change if the body rotates).

$$T = \frac{1}{2} m \vec{v}^2 + \frac{1}{2} \sum_{ij} \Omega_i \Omega_j \tag{82}$$

Comments

1. I_{ij} has 2 indices $\rightarrow$ compare with vectors (r_i, Ω_i, etc) which have one. I_{ij} is called a "2nd rank tensor", and we can write the expansion for T as a matrix multiplication.

$$\sum_{i,j} \Omega_i I_{ij} \Omega_j = \begin{bmatrix} \Omega_1 & \Omega_2 & \Omega_3 \end{bmatrix} \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix} \begin{bmatrix} \Omega_1 \\ \Omega_2 \\ \Omega_3 \end{bmatrix} \tag{83}$$

   Note that from its definition, I_{ij} is a symmetric matrix. Working out the result for discrete masses:

$$I_{ij} = \sum_a \begin{bmatrix} m_a(y_a^2 + z_a^2) & -m_a x_a y_a & -m_a x_a z_a \\ -m_a x_a y_a & m_a(x_a^2 + z^2) & -m_a y_a z_a \\ -m_a x_a z_a & -m_a y_a z_a & m_a(x_a^2 + y_a^2) \end{bmatrix} \tag{84}$$

2. If we have a continuous rigid body, all of this generalizes easily. If $\rho(\vec{r})$ is the density at the point $\vec{r}$ is the density at the point $\vec{r}$ (again, in the body-fixed frame), then

$$I_{ij} = \int_{obj} \rho(\vec{r})(\vec{r}^2 \delta_{ij} - r_i r_j) d^3\vec{r}$$

$$= \begin{bmatrix} \int \rho(\vec{r})(y^2 + z^2) d^3\vec{r} & \int \rho(\vec{r}) xy d^3\vec{r} & \int \rho(\vec{r}) xz d^3\vec{r} \\ \int \rho(\vec{r}) xy d^3\vec{r} & \int \rho(\vec{r}(x^2 + z^2) d^3\vec{r} & \int \rho(\vec{r}) yz d^3\vec{r} \\ \int \rho(\vec{r}) xz d^3\vec{r} & \int \rho(\vec{r}) yz d^3\vec{r} & \int \rho(\vec{r})(x^2 + y^2) d^3\vec{r} \end{bmatrix}$$

3. I_{ij} depends on our choice of axes $\hat{x_1}$,$\hat{x_2}$, $\hat{x_3}$

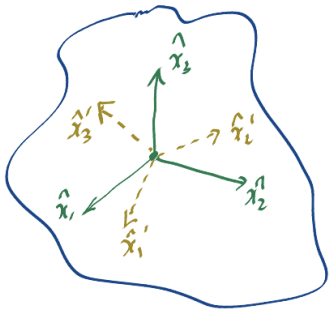


Figure 32: Orientation Axis

Claim: For any object, there always exists (at least) one set of orthogonal axes for which I_{ij} is diagonal,

$$I = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \tag{85}$$

These are known as "principle axes" and the I_i's are the "principle moments of inertia."

Proof. Recall from linear algebra that a symmetric matrix can always be diagonalized by an orthogonal transformation

$$I^{diag} = OIO^T \tag{86}$$

The matrix O transforms (rotates) you from the xyz basis used to define I_{ij} to a rotated set of axes where I is diagonal, such that $\hat{x}$', $\hat{y}$', $\hat{z}$' are principle axes (and are the eigenvectors of O).
Recall that a matrix is diagonal in the basis of its eigenvectors, we can always find the principle axes/principle moments of inertia by diagonalizing I_{ij}, but usually the principle axes are obvious from the physics. To see why, let's derive another property of I_{ij}:
Consider the angular momentum of a spinning object:

$$\vec{M} = \sum_a (\vec{r_a} \times \vec{p_a}) = \sum_a m_a (\vec{r_a} \times \vec{v_a})$$

Let's work in either the com frame of a frame where the origin is an instantaneous axis of rotation. In either case,

$$\vec{v_a} = \vec{\Omega} \times \vec{r_a}$$

and from the triple product identity, we have

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b}) \tag{87}$$

So we get

$$\vec{r_a} \times \vec{v_a} = \vec{r_a} \times (\vec{\Omega} \times \vec{r_a}) = \vec{\Omega} r^2 - \vec{r_a} \vec{Omega} \cdot \vec{r_a}$$

and so

$$\begin{aligned} M_i &= \sum_j \Omega_j \Big(\sum_a m_a (\vec{r_a}^2 \delta_{ij} - r_{ai} r_{aj}) \Big) \\ &= \sum_j I_{ij} \Omega_j \end{aligned} \tag{88}$$

or, in matrix form,

$$\begin{bmatrix} M_1 \\ M_2 \\ M_3 \end{bmatrix} = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix} \begin{bmatrix} \Omega_1 \\ \Omega_2 \\ \Omega_3 \end{bmatrix} \tag{89}$$

□

So in general, as we discussed, $\vec{M}$ & $\vec{\Omega}$ are not parallel ⇒ an object won't spin freely about an arbitrary axis, because if $\vec{\Omega}$ is constant $\vec{M}$ will rotate as it spins (since body-fixed axis rotate!)
∴ we need an external torque to change $\vec{M}$!

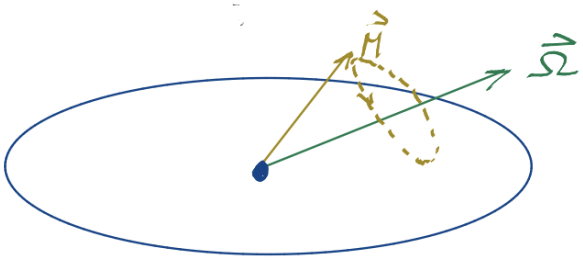


Figure 33: Spinning $\vec{M}$

But what if we choose our $\hat{x_i}$ to be principal axes?
Then we have 3 equations:

$$\begin{aligned} M_1 &= I_1\Omega_1 \\ M_2 &= I_2\Omega_2 \\ M_3 &= I_3\Omega_3 \end{aligned} \tag{90}$$

I's are not equal, but if we spin the object around a principal axis (ex. $\Omega_1 = \Omega_1$, $\Omega_2 = \Omega_3 = 0$) Then $\vec{M}$ & $\vec{\Omega}$ are parallel and $Vec\Omega$ are parallel and $\vec{M}$ stays constant as the object rotates.

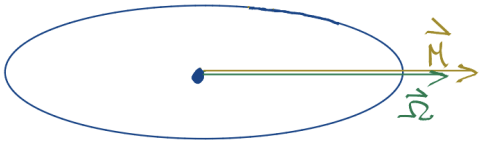


Figure 34: $\vec{M}$ & $\vec{\Omega}$

An object will spin freely ($\vec{M}$ constant) about a principal axis! So usually you can use your intuition to figure out principal axes.

Example 4 Massless Rod

Consider the following scenario:



Figure 35: Rigid Massless Rod

Here is an unfortunate (not principal) choice of axes:

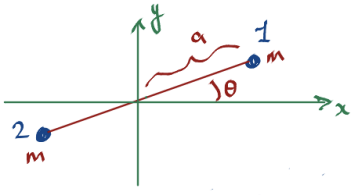


Figure 36: Rigid Massless Rod

So about these axes, $I_{11} = m(a^2 - a^2\cos^2\theta) + m(a^2 - a^2\cos^2\theta) = 2ma^2\sin^2\theta$

$$\begin{aligned} I_{12} = I_{21} &= -2ma^2\cos\theta\sin\theta \\ I_{22} &= 2ma^2(1-\sin^2\theta) = 2ma^2\cos^2\theta \\ I_{13} = I_{23} &= 0 \\ I_{33} &= 2ma^2 \end{aligned}$$

$$\therefore I = 2ma^2 \begin{bmatrix} \sin^2\theta & -\sin\theta\cos\theta & 0 \\ -\sin\theta\cos\theta & \cos^2\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{91}$$

What about the principal axes? Clearly if we choice $\theta = 0$ we'll have $I = 2ma^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, but let's solve this more formally:
we want to diagonalize I $\Rightarrow$ switch to a basis where I is diagonal, & its components are the eigenvalues of I.
Eigenvalues of I are $2ma^2$ (eigenvalue of $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$)) and $2ma^2\lambda$, where

$$\begin{bmatrix} \sin^2\theta - \lambda & -\sin\theta\cos\theta \\ -\sin\theta\cos\theta & \cos^2\theta - \lambda \end{bmatrix} = (\sin^2\theta - \lambda)(\cos^2\theta - \lambda) - \sin^2\theta\cos^2\theta = 0$$
$$\Rightarrow \lambda^2 - \lambda(\sin^2\theta + \cos^2\theta) = \lambda(\lambda - 1) = 0$$
$$\therefore \lambda = 0 \text{ or } \lambda = 1$$

So there exists a coordinate system where

$$I' = 2ma^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

We could use the eigenvectors of I to work out the change of basis but it's obvious from the physics: choosing x' to lie along the rod, we have:

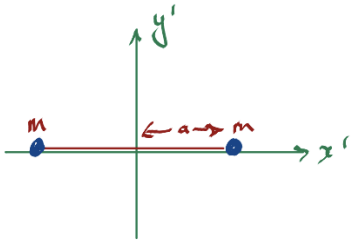


Figure 37: Rod and Principal Axis

$$I_1' \equiv I_{11}' = 0$$
$$I_2' \equiv I_2 2' = I_3' = I_{33}' = 2ma^2$$
$$I' = 2ma^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So if we define I with respect to principal axes,

$$T_{rotation} = \frac{1}{2}(I_1\Omega_1^2 + I_2\Omega_2^2 + I_3\Omega_3^2) \tag{92}$$

General observations about principal axes

If an object has some degree of symmetry, principal axes are usually obvious

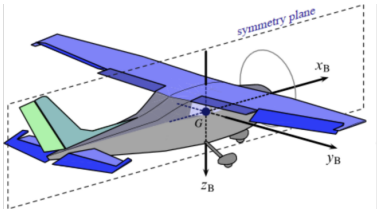


Figure 38: Symmetry Plane

1. plane of symmetry $\Rightarrow$ one principal axis is $\perp$ to the plane, the other 2 are in the plane (& COM is in the plane)
2. axis of symmetry: one principal axis is along the axis, the other 2 are $\perp$ to it (and by symmetry, have the same I_i)

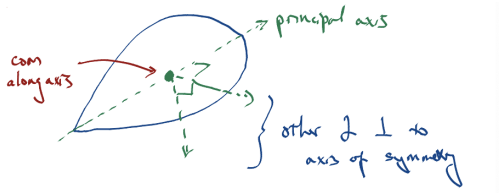


Figure 39: Axis of symmetry

If $I_1 = I_2 = I_3$, the object is called a "spherical top" and any set of axis through the COM are principal axes, and since $I_1 = I_2 = I_3$, $\vec{M}$ and $\vec{\Omega}$ are parallel, we can spin a spherical top about any axis passing through the COM. If $I_1 = I_2 \neq I_3$, the object is called a "symmetrical top":

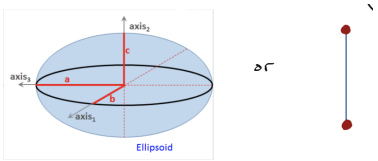


Figure 40: Symmetrical Top

And if $I_1 \neq I_2 \neq I_3$ the object is called an "asymmetrical top". Note that for any object,

$$I_1 + I_2 = \sum_a m_a(x_a^2 + z_a^2 + y_a^2 + z_a^{2\prime})$$

$$\sum_a m_a(x_a^2 + y_a^2 + 2z_a^2) \geq \sum_a m_a(x_a^2 + y_a^2) = I_3$$

Therefore none of the principal moments of inertia can exceed the sum of the other two. Finally, I_{ij} depends on the origin O. If we know I_{ij} for a set of axes passing through the COM, what is I'_{ij} for a parallel set of axis passing through a point O' displaced by $\vec{b}$?

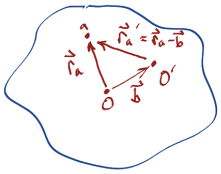


Figure 41: Parallel Axes

$$I'_{ij} = \sum_a m_a(\delta_{ij} r'^2_a - r'_{ai} r_{aj})$$
$$= \sum_a m_a(\delta_{ij}(r_a^2 - 2\vec{b}\cdot\vec{r_a} + b^2) - r_{ai}r_{aj} + r_{ai}b_j + b_i r_{aj} - b_i b_j)$$

but for COM, $\sum_a m_a \vec{r_a} = 0$, $\therefore 2\vec{b}\cdot\vec{r_a}$, $r_{ai}b_j$, $b_i r_{aj}$ vanish in the sum.

$$\therefore I'_{ij} = \sum_a m_a(\delta_{ij} r_a^2 - r_{ai}r_{aj}) + \sum_a m_a(b^2\delta_{ij} - b_i b_j) \tag{93}$$

$$I'_{ij} = I_{ij} + \mu(b^2\delta_{ij} - b_i b_j) \text{ Parallel Axis Theorem} \tag{94}$$

I_{ij} for a solid cube

A cube is a "spherical top" so we can choose any set of principal axes through the COM...easiest are those going through the face:

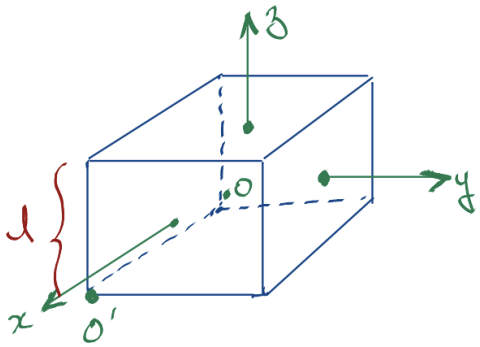


Figure 42: Solid Cube Example

We have

$$I_{11} = \int_{-l/2}^{l/2}\int_{-l/2}^{l/2}\int_{-l/2}^{l/2} \rho(y^2+z^2)dxdydz = \frac{Ml^2}{6}$$

$$\therefore I_{ij} = \frac{Ml^2}{6}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} \tag{95}$$

What about I' measured about O'? We can just translate axes by

$$\vec{b} = -\frac{l}{2}\hat{x} - \frac{l}{2}\hat{y} - \frac{l}{2}\hat{z}$$

$$\therefore b^2 = \frac{3l^2}{4} \text{ , } b_i b_j \frac{l^2}{4} \text{ for any i,j}$$

Therefore

$$I'_{ij} = \frac{Ml^2}{6}\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix} + \frac{3Ml^2}{4}\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix} - \frac{l^2}{4}\begin{bmatrix}1&1&1\\1&1&1\\1&1&1\end{bmatrix}$$

$$= Ml^2\begin{bmatrix}\frac{2}{3}&-\frac{1}{4}&-\frac{1}{4}\\-\frac{1}{4}&\frac{2}{3}&-\frac{1}{4}\\-\frac{1}{4}&-\frac{1}{4}&\frac{2}{3}\end{bmatrix} \Rightarrow \text{ Not principal axes} \tag{96}$$

Rotations about edges of circle do not have$\vec{M} \parallel \vec{\Omega}$

Exercise: What are the principal axis about O'? Can you show this from the parallel axis theorem?

Free Rotations of Rigid Bodies

Before we consider the addition of external forces/torque. Let's consider the free rotation of a rigid body in the COM frame (so $\vec{v} = 0$). Then

$$L = T - \frac{1}{2}\sum_{ij} \Omega_i I_{ij} \Omega_j \tag{97}$$

Since $\Omega_i = \frac{d\theta_i}{dt}$ are the generalized velocities, so

$$\frac{d}{dt}(\frac{\partial L}{\partial \Omega_k}) = 0 \Rightarrow \frac{d}{dt}(\frac{1}{2}\sum_{ij}\left[\frac{\partial \Omega_i}{\partial \Omega_k} I_{ij}\Omega_j + \Omega_i I_{ij}\frac{\Omega_j}{\partial \Omega_k}\right])$$

$$= \frac{d}{dt}\left[\sum_j I_{kj}\Omega_j\right] = \frac{d}{dt}M_k = 0 \tag{98}$$

Unsurprisingly, we get $\frac{d\vec{M}}{dt} = 0$ for free rotation. If there are external forces, we have:

$$\frac{d}{dt}\vec{M} = \frac{d}{dt}\sum_a \vec{r_a} \times \vec{p_a} = \sum_a \vec{r_a} \times \vec{p_a} + \sum_a \vec{r_a} \times \vec{F_a}$$

$$\Rightarrow \frac{d\vec{M}}{dt} = \vec{N} \tag{99}$$

Example 5 Spherical Top

So the dynamics of a free spherical top are pretty trivial...any axis is a principal axis, so defining $\hat{x_3}$ to be the rotation axis, we have

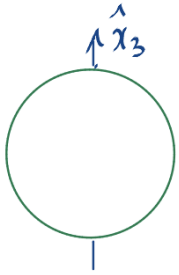


Figure 43: Rotation Axis

We have $M_3 = I\Omega_3 =$ constant and $M_{1,2} = 0$. Therefore the object spins at a constant $\Omega = \Omega_3$

Example 6 Symmetrical Top

Recall we have $I_1 = I_2 \neq I_3$. If we rotate about a principal axis, it's the same as the spherical top:

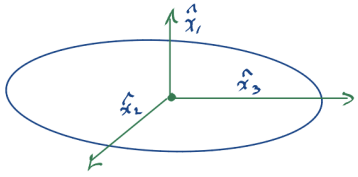


Figure 44: Rotation Axis

We have $M_i = I_i \Omega_i$ = constant (of $i = 1\,2$ or 3) But what happens if the axis of rotation is not along a principal axis? — Then it wobbles as it spins. Let's analyze this now! Consider the following system:

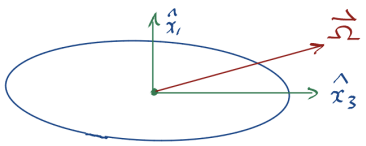


Figure 45: Wobbling System

Despite the wobbling, $\vec{M}$ is constant. At this instant, take $\vec{\Omega}$ to be in the $x_1 - x_3$ plane, as in the figure, we have $\Omega_2 = 0 \Rightarrow M_2 = 0$. In this case, $I_1 > I_3$ because $\vec{M}$ points as indicated here:

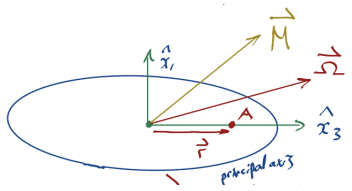


Figure 46: Orientation of $\vec{M}$

So $\vec{M}$, axis of rotation Ω and $\hat{x_3}$ all lie in a plane, denoted as the plane S. As the ball rotates, S will rotate, but the three vectors will continue to lie in a plane. Let's now consider the motion of the point A on the figure, on the $\hat{x_3}$ axis (which is a body-fixed axis: it rotates with the ball). Then the velocity of A is:

$$\vec{v_A} = \vec{\Omega} \times \vec{r_A} \Rightarrow \text{ into the page, } \perp \text{ to} \vec{M}\,\vec{\Omega}\,\hat{x_3} \text{ and S} \tag{100}$$

So at every instant the axis of the top moves $\perp$ to the plane S. But since $\vec{M}$ is fixed, $\hat{x_3}$ must rotate uniformly about $\vec{M}$, as does the axis of rotation $\vec{\Omega}$. Note: the object is rotating about Ω, while Ω is rotating about $\vec{M}$! Therefore $\hat{x_3}$ & $\vec{\Omega}$ follow the dashed curves as they rotate about fixed $\vec{M}$:

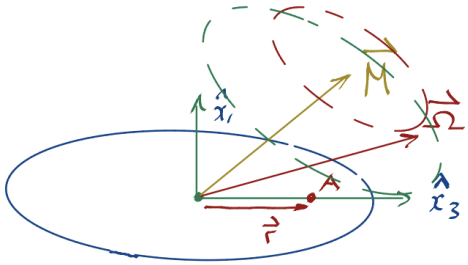


Figure 47: More Illustration on Rotations

Therefore the symmetrical top precesses about the axis $\vec{M}$ "Torque-free precession"/"regular precession" $\Rightarrow$ this is the technical term for the wobble.

Rate of Precession/Angular Velocity of $\vec{\Omega}$ (or $\hat{x_3}$) about $\vec{M}$

We are going to develop 2 different formalisms to tackle this problem because there are 2 "natural" ways to think about it. They will give the same answer but from 2 different perspectives.

1. The Euler Equations are defined using the body-fixed coordinates system (which is rotating with the object). They are simple because we choose the axes of our coordinate system to be the principal axes of the rotating object.
   Disadvantage: All of the results will be in a rotating reference frame, which can make the results tricky to visualize in an inertial frame.
   For example, for the precessing football, $\vec{\Omega}$ is changing direction, and the $\hat{x_i}$ axes are rotating with the ball. Therefore, the components of Ω_i change for both reasons and it's nontrivial to disentangle them.

2. The Euler Angles are defined in the lab/inertial frame. Since we are typically interested in what a spinning object looks like in an inertial frame, not a frame spinning with the object, this approach is easier to interpret.
   They are based on a simple way to describe the rotational motion of an external object.
   Consider the following system

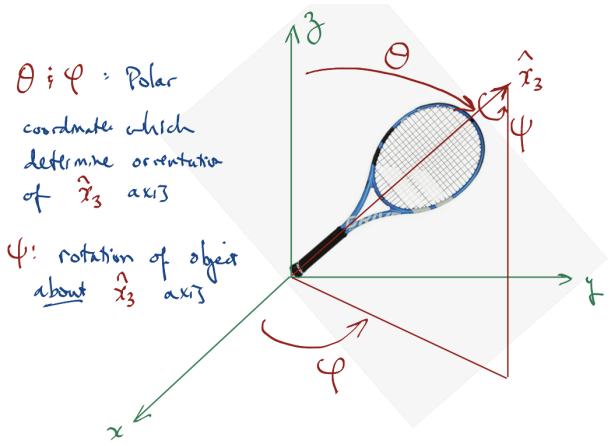


Figure 48: Spinning Tennis Racket

   In the lab frame, it's more convenient to describe the motion of the object with $\dot{\theta}$, $\dot{\varphi}$, $\dot{\psi}$ rather than Ω_1 Ω_2, Ω_3 (defined w.r.t rotating principal axes).

(1) Euler Equations

$$\frac{d\vec{M}}{dt} = 0 = \sum_i \frac{d}{dt}\left[M_i(t)\hat{x_i}(t)\right]$$
$$= \sum_i \left[\frac{dM_i}{dt}\hat{x_i} + M_i\frac{d\hat{x_i}}{dt}\right]$$

But we know that for any fixed vector $\vec{a}$ on the body,

$$\frac{d\vec{a}}{dt} = \vec{\Omega} \times \vec{a} \tag{101}$$

Therefore, the basis vectors (body-fixed) obey

$$\frac{d\hat{x_i}(t}{dt} = \vec{\Omega} \times \hat{x_i} \text{ (rotating)}$$

So we have

$$\frac{d\vec{M}}{dt} = \sum_i \left[\frac{dM_i}{dt}\hat{x_i} + M_i(\vec{\Omega} \times \hat{x_i}) \right] = 0 \tag{102}$$

let's look at the $\hat{x_1}$ component of this. Recall that

$$(\vec{a} \times \vec{b})_a = a_2 b_3 - a_3 b_2 \tag{103}$$

$$\therefore (\vec{\Omega} \times \hat{x_2})_1 = -\Omega_3 \;\; (\vec{Omega} \times \hat{x_3})_1 = \Omega_2 \, (\Omega \times \hat{x_1})_1 = 0$$

and so we have

$$\frac{dM_1}{dt} - M_2\Omega_3 + M_3\Omega_2 = 0 \tag{104}$$

Now, since $\hat{x_i}$ are principal axes, we know $M_i = I_i\Omega_i$, $i = 1...3$, so this becomes

$$I_1\dot{\Omega}_1 + \Omega_2\Omega_3(I_3 - I_2) = 0 \tag{105}$$

Doing the same analysis for $\hat{x_2}$ & $\hat{x_3}$, we obtain the results:

Euler Equations (for free rotation)

$$\begin{aligned} I_1\dot{\Omega}_1 + \Omega_2\Omega_3(I_3 - I_2) &= 0 \\ I_2\dot{\Omega}_2 + \Omega_3\Omega_1(I_1 - I_3) &= 0 \\ I_3\dot{\Omega}_3 + \Omega_1\Omega_2(I_2 - I_1) &= 0 \end{aligned} \tag{106}$$

note: if $\vec{N} \neq 0$, the RHS of these equations is N_1, N_2, N_3, respectively. Examples:

1. Spherical top: $I_1 = I_2 = I_3$

   $$\therefore \dot{\Omega}_1 = \dot{\Omega}_2 = \dot{\Omega}_3 = 0 \Rightarrow \vec{\Omega} \text{ is constant} \tag{107}$$

   spins freely about any axis

2. Symmetrical top: $I_! = I_2 \neq I_3$

   $$\begin{aligned} \dot{\Omega}_1 &= \Omega_2\Omega_3(\frac{I_1 - I_2}{I_1}) \\ \dot{\Omega}_2 &= -\Omega_2\Omega_3(\frac{I_1 - I_3}{I_1}) \\ \dot{\Omega}_3 &= 0 \Rightarrow \Omega_3 = \text{ constant} \end{aligned} \tag{108}$$

   So if $\Omega_1 = \Omega_2 = 0$ initially, $\dot{\Omega}_1 = \dot{\Omega}_2 = 0$ so they stay 0, and the top spins freely about the $\hat{x_3}$ axis, but otherwise Ω is not in the $\hat{x_3}$ direction. Ω_3 is still constant, but

   $$\begin{aligned} \dot{\Omega}_1 &= -\omega\Omega_2 \\ \dot{\Omega}_2 &= \omega\Omega_1 \end{aligned} \quad \text{where } \; \omega \equiv \Omega_3(\frac{I_3 - I_1}{I_1}) = \text{ constant} \tag{109}$$

   Therefore we have

   $$\begin{aligned} \ddot{\Omega}_1 &= -\omega\dot{\Omega} = -\omega^2\Omega_1 \\ \ddot{\Omega}_2 &= -\omega\dot{\Omega}_1 = -\omega^2\Omega_2 \end{aligned} \tag{110}$$

   So Ω_1 & Ω_2 oscillate with frequency ω

   $$\begin{aligned} \Omega_1 &= \Omega_0 \cos\omega t \\ \Omega_2 &= \Omega_0 \sin\omega t \end{aligned} \tag{111}$$

Just like the following figure, Ω precesses about the $\hat{x_3}$ axis with frequency ω.

if $I_3 > I_1$ (object is short and wide like a coin), then $\omega > 0$.

If $I_1 > I_3$ (object is long and thin like a football), then $\omega < 0$ (so $\vec{\Omega}$ rotates reversely).

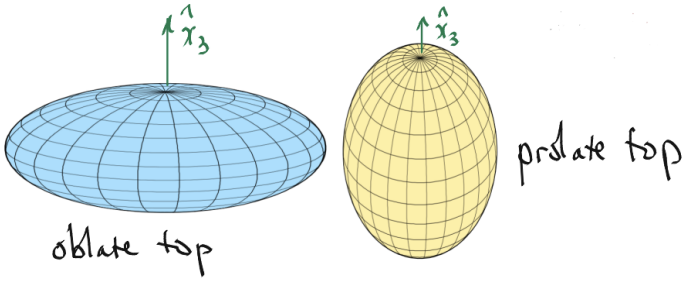


Figure 49: Oblate top & Prolate top

Therefore, in the body-fixed frame, Ω precesses about the $\hat{x_3}$ axis with constant frequency $\omega = \Omega_3 \left(\frac{I_3 - I_1}{I_1}\right)$

Example 7 Chandler Wobble

Earth is not a perfect sphere due to equatorial bulge

$$\frac{I_3 - I_1}{I_1} \quad \frac{1}{300}$$

Axis of rotation $\vec{\Omega}$ is not precisely aligned with principal axis $\hat{x_3} \Rightarrow$ axis of rotation precesses about $\hat{x_3}$

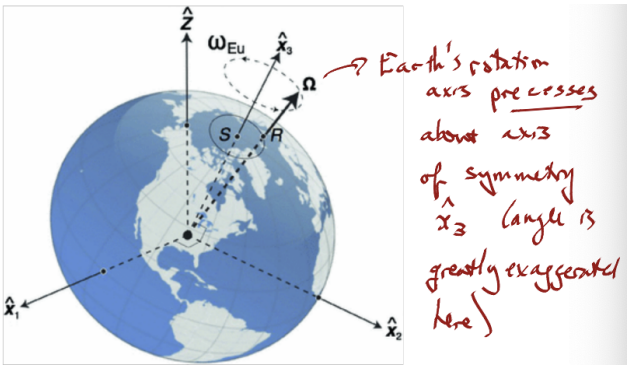


Figure 50: Earth Wobble

Since $\Omega_3 = \frac{2\pi}{day}$, $\omega \approx \frac{2\pi}{300}(\text{day})^{-1}$ Therefore the Earth should precess with a period of approximately 300 days. In reality, the period is actually 427 days because Earth is not completely rigid.

Example 8 Wobbly Football

Consider the following system:

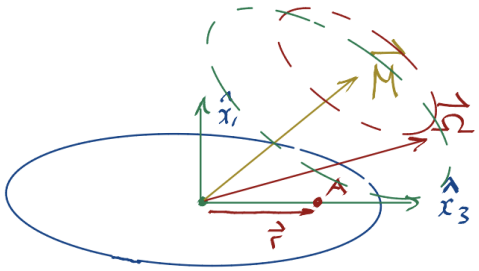


Figure 51: Wobbly Football

ω is the precession frequency of Ω about $\hat{x_3}$ in the rotating (body-fixed) frame. In the rotating frame, $\hat{x_3}$, $\vec{\Omega}$, $\vec{M}$ lie in a plane S which is precessing about $\hat{x_3} with frequency \omega$

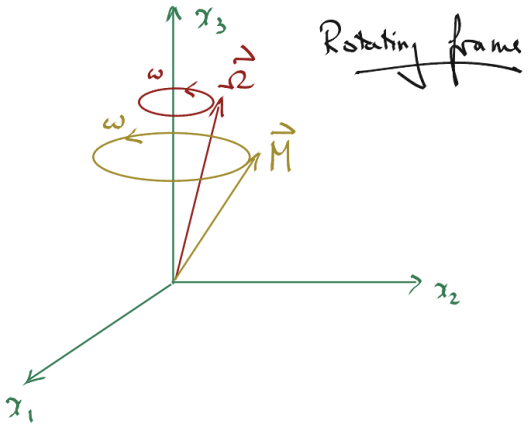


Figure 52: Rotating Frame

The plane S is rotating at $\omega\hat{x_3}$ in a frame which is rotating at $\vec{\Omega}$ in the lab frame Therefore, the plane S is rotating at $\vec{\Omega} + \omega\hat{x_3}$ in the lab frame. At the instant that S is in the $x_1 - x_3$ plane,

$$\begin{aligned}
\vec{\Omega} + \omega\hat{x_3} &= \Omega_1\hat{x_1} + \Omega_3\hat{x_3} + \Omega_3(\frac{I_3 - I_1}{I_1})\hat{x_3} \\
&= \Omega_1\hat{x_1} + \Omega_3\hat{x_3} + \Omega_3(\frac{I_3}{I_1} - 1)\hat{x_3} \\
&= \Omega_1\hat{x_1} + \frac{I_3}{I_1}\Omega_3\hat{x_3} \\
&= \frac{1}{I_1}(I_1\Omega_1\hat{x_1} + I_3\Omega_3\hat{x_3}) \\
&= \frac{\vec{M}}{I_1}
\end{aligned}$$

Therefore in the lab frame, the top precesses with angular velocity

$$\vec{\Omega_{pr}} = \frac{\vec{M}}{I_1} \tag{112}$$

So as we see, the rotation between $\vec{Omega}$, ω & Ω_{pr} is not trivial! Before we solve this problem in a completely different way using the Euler angles, here's another nice application of the Euler angles, here's another nice application of the Euler equations, this time to an asymmetrical top.

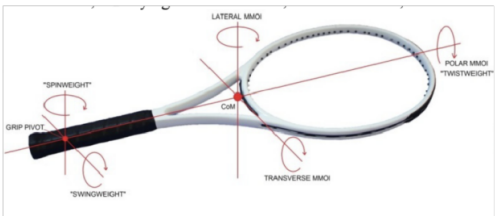


Figure 53: Tennis Racket

More on Asymmetrical Top

We know an asymmetrical top spins freely about any of its principal axes, but what if we are imperfect and don't spin it precisely about a principal axis? Is rotation about a principal axis stable?
Consider adding a small perturbation to rotation about a principal axis, say $\hat{x_i}$:

$$\begin{aligned} \Omega_1 = \Omega \text{ constant} &\Rightarrow \Omega_1 = \Omega + \eta_1(t) \\ \Omega_2 = 0 &\Rightarrow \Omega_2 = \eta_2(t) \\ \Omega_3 = 0 &\Rightarrow \Omega_3 = \eta_3(t) \end{aligned} \tag{113}$$

Since the η_i's are small perturbations $\frac{\eta_i}{\Omega} << 1$, we can substitute them into the Euler equations and neglect terms of $O(\eta^2)$, which gives

$$\begin{aligned} I_1\dot{\eta_1} &= -\eta_2\eta_3(I_3 - I_2) = O(\eta^2) \\ \therefore I_1\dot{\eta_1} &\approx 0 \\ \text{and } I_2\dot{\eta_2} &\approx \Omega\eta_3(I_3 - I_1) \\ I_3\dot{\eta_3} &\approx \Omega\eta_2(I_1 - I_2) \end{aligned}$$

Taking $\frac{d}{dt}$ of equation for $\dot{eta_2}$ and substituting the equation for $\dot{\eta_3}$, we get

$$\begin{aligned} I_2\ddot{\eta_2} &= \frac{\Omega^2}{I_3}(I_3 - I_1)(I_1 - I_2)\eta_2 \\ &\equiv A\eta_2 \quad \text{where} \quad A \equiv \Omega^2\frac{(I_3 - I_1)(I_1 - I_2)}{I_3} \end{aligned}$$

Note if $A < 0$, the solution is of the form $\eta_2 \; a \sin \alpha t + b \cos \alpha t$. Therefore the perturbation does not grow endlessly $\rightarrow$ it is stable.

But if $A > 0$, solution is of the form $\eta_2 \; exp\alpha t (+exp - \alpha t)$. Therefore the perturbation grows exponentially $\rightarrow$ unstable!

$\therefore$ rotation about principal axis $\hat{x_1}$ is unstable if

$$\begin{gathered} I_2 < I_1 < I_3 \\ \text{or} \\ I_2 > I_1 > I_3 \end{gathered} \tag{114}$$

Rotation about a principal axis is stable for axes with larger and smallest I but unstable about the intermediate axis!

This is known variously as the "intermediate axis theorem", the "tennis racket theorem", or the "Dzanibekov Effect". To illustrate it, try to flip a tennis racket about all 3 principal axes.

(2) Euler Angles

Let's switch to working in the lab/inertial frame and define the Euler Angles θ, φ, ψ. The tricky part is going to be relating $\dot{\theta}$, $\dot{\varphi}$, $\dot{\psi}$ to the components of $\vec{\Omega}$ in the body-fixed frame. Let's figure out how to relate the body fixed axes $\hat{x_i}$ to the lab axes X, Y, Z by a series of rotations.

To begin, recall the form of a rotation in 2D:

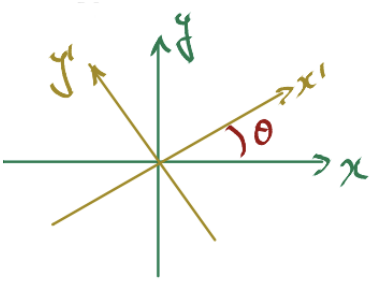


Figure 54: Rotation in 2D

$$\begin{aligned} \hat{x}' &= \cos\theta \hat{x} + \sin\theta \hat{y} \\ \hat{y}' &= -\sin\theta \hat{x} + \cos\theta \hat{y} \end{aligned} \tag{115}$$

Or in matrix form

$$\begin{bmatrix} \hat{x}' \\ \hat{y}' \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix} \tag{116}$$

The matrix form is convenient became if we want to do 2 successive rotations by θ_1 & θ_2 it just becomes matrix multiplication

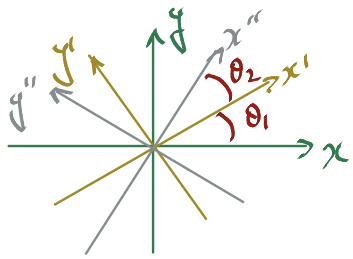


Figure 55: Rotating Axis in 2D

$$\begin{bmatrix} \hat{x}'' \\ \hat{y}'' \end{bmatrix} = R(\theta_2) \begin{bmatrix} \hat{x}' \\ \hat{y}' \end{bmatrix} = R(\theta_2)R(\theta_1) \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix} \tag{117}$$

$$= R(\theta_1 + \theta_2) \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix} = \begin{bmatrix} \cos(\theta_1 + \theta_2) & \sin(\theta_1 + \theta_2) \\ -\sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix} \tag{118}$$

Therefore, to apply successive rotations, we just need to do matrix multiplication!

This works in 3D as well. In particular, from the form of a 2D rotation, it's obvious that a 3D rotation about the z axis takes the form

$$R_z(\theta) \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{119}$$

and a rotation about the x axis is

$$R_x(\theta) \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} \tag{120}$$

So now we can figure out the relation between the body-fixed x_i axes and the lab X_i axes in terms of the Euler angles in 3 steps:

1. Rotate by an angle *varphi* about the z axis to x_i' axes:

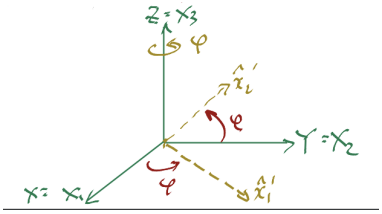


Figure 56: Rotation Step 1

$$\hat{x_i}' = \sum_j R_3(\varphi)_{ij} \hat{X}_j \tag{121}$$

2. Rotate by θ about the $\hat{x_1}'$ axis

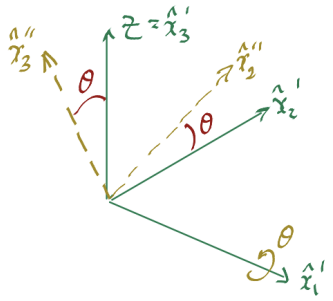


Figure 57: Rotation Step 2

$$\hat{x_i}'' = \sum_j R_1(\theta)_{ij} \hat{x_j}' \tag{122}$$

3. Finally, rotate by ψ about the $\hat{x_3}''$ axis to get the body-fixed axes.

$$\hat{x_i} = \sum_j R_3(\psi)_{ij} \hat{x_j}'' \tag{123}$$

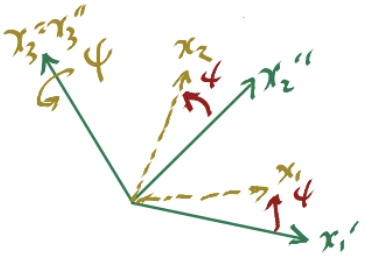


Figure 58: Rotation Step 3

There are the 3 rotations we described earlier! But now we can see our rotation matrices to work out the precise relation between the axes

$$\begin{bmatrix} \hat{x_1} \\ \hat{x_2} \\ \hat{x_3} \end{bmatrix} = R_3(\psi) R_1(\theta) R_3(\varphi) \begin{bmatrix} \hat{X} \\ \hat{Y} \\ \hat{Z} \end{bmatrix} \tag{124}$$

$$\therefore R(\theta\, \varphi, \psi) = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{125}$$

$$= \begin{bmatrix} \cos\varphi\cos\varphi - \cos\theta\sin\varphi\sin\psi & \sin\varphi\cos\psi + \cos\theta\sin\psi\cos\varphi & \sin\theta\sin\psi \\ -\cos\varphi\sin\psi - \cos\theta\cos\psi\sin\varphi & -\sin\psi\sin\varphi + \cos\theta\cos\psi\cos\varphi & \sin\theta\cos\psi \\ \sin\theta\sin\varphi & -\sin\theta\cos\varphi & \cos\theta \end{bmatrix} \tag{126}$$

$\hat{x_1}$, $\hat{x_2}$, $\hat{x_3}$ in terms of 3 Euler angles.

Now...to get the components of $\vec{\Omega}$ in terms of $\dot{\theta}$, $\dot{\varphi}$, $\dot{\psi}$, we write

$$\hat{x_i}(t) = \sum_j R_{ij}(\psi_1, \theta, \varphi)\hat{X}_j \tag{127}$$

$$\therefore \frac{d\hat{x_i}(t)}{dt} = \sum_j \frac{dR_{ij}(t)}{dt}\hat{X}_j = \sum_{jk} \frac{dR_{ij}}{dt}(R^{-1})_{jk}\hat{x_k} \tag{128}$$

But we know in general $\frac{d\hat{x_i}}{dt} = (\vec{\Omega} \times \hat{x_i})$
So given R^{-1} & $\frac{dR}{dt}$ (depend on θ, φ, ψ, $\dot{\theta}$, $\dot{\varphi}$, $\dot{\psi}$) we can solve

$$\sum_j \frac{dR_{ij}}{dt}(R^{-1})_{jk}\hat{x_j} = (\vec{\Omega} \times \vec{x})_i \text{ for the } \Omega_i\text{'s} \tag{129}$$

In terms of the Euler angles.

Rather than work through all that, we can get the answer by thinking about this physically (consider an infinitesimal rotation)

$$(\psi\, \theta\, \varphi) \to (\psi + d\psi\, \theta + d\theta\, \varphi + d\varphi) \text{ in a time} dt \tag{130}$$

Then, going back to our original construction, the instantaneous angular velocity is

$$\vec{\Omega} = \dot{\varphi}\hat{z} + \dot{\theta}\hat{x_1}' + \dot{\psi}\hat{x_3} \tag{131}$$

So we just need to write $\hat{z}$ & $\hat{x_1}$ in the body-fixed basis. This is straightforward:

1. $$\begin{bmatrix} \hat{X} \\ \hat{Y} \\ \hat{Z} \end{bmatrix} = R^{-1}(\psi\, \theta\, \varphi) \begin{bmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{bmatrix} \tag{132}$$

   And rather than work out R^{-1}_{ij} by hand, we note that we invert the rotation R by undoing the stpes in the reverse order:

   $$\begin{aligned} R^{-1}(\psi, \theta, \varphi) &= R_3(-\varphi)R_1(-\theta)R_3(-\psi) \\ &= R(-\varphi\ -\theta\ -\psi) \end{aligned}$$

   Therefore, to invert the matrix R, change the angles to the opposite sign and then swap ψ & φ

2. Similarly, we just have to rotate $\hat{x_1}$ by $-\psi$ about the $\hat{x_3}$ axis to get $\hat{x_1}'$:

   $$\hat{x_1}' = \cos\psi\hat{x_1} - \sin\psi\hat{x_2} \tag{133}$$

   Therefore, substituting the expressions for $\hat{x_1}$ and $\hat{z}$ into the expression for $\vec{\Omega}$ and taking the $\hat{x}_{1,2,3}$ components gives

   $$\begin{aligned} \Omega_1 &= \dot{\varphi}\sin\theta\sin\psi + \dot{\theta}\cos\psi \\ \Omega_2 &= \dot{\varphi}\sin\theta\cos\psi - \dot{\theta}\sin\psi \\ \Omega_3 &= \dot{\psi} + \dot{\varphi}\cos\theta \end{aligned} \tag{134}$$

Let's check that these equations make sense:

1. $\theta(t) = 0$: In this case, $\hat{z} = \hat{x_3}$ and so φ (rotation about $\hat{z}$) and $\psi(...\hat{x_3})$ are the same rotation. Our formula gives
$$\begin{aligned} \Omega_1 = \Omega_2 = 0 \\ \Omega_3 = \dot{\psi} + \dot{\varphi} \end{aligned} \tag{135}$$

2. $\theta(t) = \frac{\pi}{2}$: In this case, the $\hat{x_3}$ axis is perpendicular to the $\hat{z}$ axis. the object is spinning about the $\hat{x_3}$ axis at $\dot{\psi}$ and the $\hat{x_3}$ axis is precessing at $\dot{\varphi}$ in the $\hat{x_1} - \hat{x_2}$ plane:

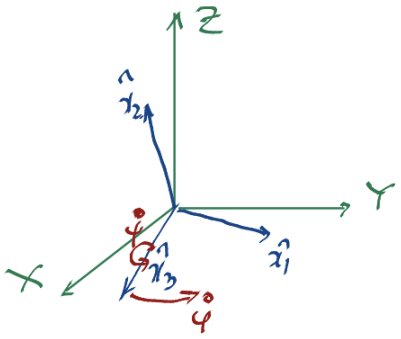


Figure 59: Precession Demonstration

$$\begin{aligned} \Omega_3 &= \dot{\psi} \\ \Omega_1 &= \dot{\varphi} \sin\psi \\ \Omega_2 &= \dot{\varphi} \cos\psi \end{aligned} \tag{136}$$

So let's look for a third time at the symmetrical top, this time using Euler angles:

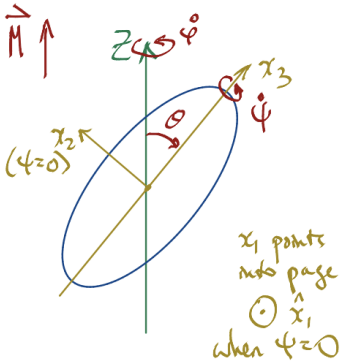


Figure 60: Euler Angles for Symmetrical Top

Choose our $\hat{X}$, $\hat{Y}$, $\hat{Z}$ coordinates so that $\vec{M}$ is in the $\hat{z}$ direction (constant). Recall:

$$\begin{aligned} M_1 &= I_1\Omega_1 \\ M_2 = I_2\Omega_2 &= I_1\Omega_2 \\ M_3 &= I_3\Omega_3 \end{aligned} \tag{137}$$

Let's draw the axis at the instant $\psi = 0$. Then $\hat{x_1}' = \hat{x_1}$ and so rotation θ is about the $\hat{x_1}$ axis. Therefore $\hat{x_1} \perp$ plane containing $\vec{M}$ & $\hat{x_3}$ (into page) and from our general result we have

$$\Omega_1 = \dot{\theta} \quad \Omega_2 = \dot{\varphi} \sin\theta \quad \Omega_3 = \dot{\varphi} \cos\theta + \dot{\psi} \tag{138}$$

From these equations we can immediately read off our previous results:

1. $\hat{x_1} \perp \vec{M}$ Therefore $M_1 = 0 \Rightarrow \Omega_1 = 0 \Rightarrow \dot{\theta} = 0$. So θ is contour. The angle between $\vec{M}$ & $\hat{x_3}$ is fixed. so $\hat{x_3}$ rotates about $\vec{M}$.

2. $\dot{\varphi}$ is the frequency with which $\hat{x_3}$ rotates about $\hat{z}$. Therefore, $\Omega_{pr} = \dot{\varphi}$ is the precession frequency.

$$\dot{\varphi} = \frac{\Omega_2}{\sin\theta} = \frac{M_2}{I_2 \sin\theta} \text{ but } \quad M_2 = M \sin\theta \tag{139}$$

$$\therefore \dot{\varphi} = \frac{M}{I_2} = \frac{M}{I_1} \tag{140}$$

3. $\dot{\psi}$ is the angular frequency of the $\hat{x_{1,2}}$ axes about $\hat{x_3} \rightarrow$ this is what we called ω before, in the rotating frame.

$$\text{and } \Omega_3 = \frac{M_3}{I_3} = \dot{\varphi} \cos\theta + \dot{\psi} = \frac{M \cos\theta}{I_1} + \dot{\psi} \tag{141}$$

$$\text{but } M \cos\theta = M_3 \tag{142}$$

$$\therefore \dot{\psi} = \frac{M_3}{I_3} - \frac{M_3}{I_1} = \frac{M_3}{I_3}\left(\frac{I_1 - I_3}{I_1}\right) = \Omega_3\left(\frac{I_3 - I_1}{I_1}\right) = \omega \tag{143}$$

So we get the same results as the Euler equations, but an even simpler derivation.

The Heavy Symmetric Top with Gravity

Let's put this altogether with an external force/torque! Consider the motion of a spinning top with gravity:

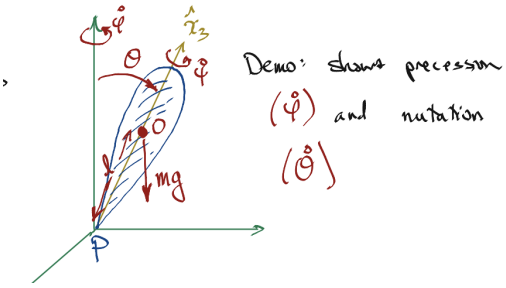


Figure 61: Spinning Top under Gravity

The Euler angles are nicely set up for this problem:

1. Top spins about its axis at $\dot{\psi}$
2. Axis of rotation precesses about z axis ($\dot{\varphi}$) and exhibits nutation ($\dot{\theta}$)

Because P is stationary, it defines an instantaneous axis of rotation. Using P as the origin, not the COM O. The Lagrangian of the top is then

$$L = \frac{1}{2}I_1(\Omega_1^2 + \Omega_2^2) + \frac{1}{2}I_3\Omega_3^2 - \mu gl\cos\theta \tag{144}$$

$$= \frac{1}{2}I_1(\dot{\theta}^2 + \sin^2\theta\dot{\varphi}^2) + \frac{1}{2}I_3(\dot{\psi} + \cos\theta\dot{\varphi})^2 - \mu gl\cos\theta \tag{145}$$

where I_i's are defined about P, not O

To analyze the top, let's first look for the conserved quantities.

1. L is independent of ψ and φ: cyclic coordinates
   $\therefore P_\psi = \frac{\partial L}{\partial\dot{\psi}}$ and $P_\varphi = \frac{\partial L}{\partial\dot{\varphi}}$ are conserved, where

$$P_\psi = \frac{\partial L}{\partial\dot{\psi}} = I_3(\dot{\psi} + \cos\theta\dot{\varphi}) = I_3\Omega_3$$

$$\therefore \Omega_3 \text{ is constant}$$

$$P_\varphi = \frac{\partial L}{\partial\dot{\varphi}} = I_1\sin^2\theta\dot{\varphi} + I_3\cos\theta(\dot{\psi} + \dot{\varphi}\cos\theta)$$

$$\therefore I_3\Omega_3\cos\theta + I_1\sin^2\theta\dot{\varphi} \text{ is conserved}$$

Note that P_ψ and P_φ are the components of angular momentum about the $\hat{x_3}$ & $\hat{z}$ axes, respectively. $\vec{M}$ is not conserved because the system isn't fully rotationally invariant, but these 2 components of $\vec{M}$ are.

2. The system is invariant under time translations $\therefore E = T + V = \frac{1}{2}I_i(\dot{\theta}^2 + \dot{\varphi}^2 \sin^2\theta) + \frac{1}{2}I_3\Omega_3^2 + Mgl\cos\theta$ is conserved.
   So 3 d.o.f, 3 conserved quantities $\Rightarrow$ the system is integrable. Therefore, we can solve just any conservation laws!

Special case $\dot{\theta} = 0$ (Pure precession, no nutation)

Then we can use $\frac{dE}{dt} = 0$ or, equivalently, the remaining E-L equation $\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = \frac{\partial L}{\partial \theta}$ to get

$$I_1\ddot{\theta} = (\mu gl + I\dot{\varphi}^2\cos\theta - I_3\Omega_3\dot{\varphi})\sin\theta \tag{146}$$

and

$$\ddot{\theta} = \dot{\theta} = 0 \Rightarrow I\dot{\varphi}^2\cos\theta - I_3\Omega_3\dot{\varphi} + \mu gl = 0 \tag{147}$$

$$\Rightarrow \dot{\varphi} = \frac{I_3\Omega_3}{2I_1\cos\theta}(1 \pm \sqrt{1 - \frac{4\mu I_1 gl\cos\theta}{I_3^2\Omega_3^2}}) \tag{148}$$

Note: If $\cos\theta > 0$ ($\theta < \pi/2$), the top is pointing upp, and there is a minimum frequency $\Omega_3 = \frac{2}{I_3}\sqrt{I_1\mu gl\cos\theta}$ below which the top will not precess with constant θ (top will drop).

For the case $I_3\Omega_3 >> \sqrt{I_1\mu gl}$, we can taylor expand the square root $\sqrt{1+a} = 1 + \frac{1}{2}a + ...$ to get

$$\dot{\varphi} \approx \frac{\mu gl}{I_3\Omega_3} \text{ (slow)} \tag{149}$$

or

$$\dot{\varphi} \approx \frac{I_3\Omega_3}{I_1\cos\theta} \text{ (fast)} \tag{150}$$

Precessing Top with Nutation ($\dot{\theta} \neq 0$)

Now, let's look at the general case. Since the system is integrable, we know

$$P_\psi = I_3\Omega_3 \tag{151}$$

$$P_\varphi = I_1\sin^2\dot{\varphi} + I_3\cos\theta(\dot{\psi} + \dot{\varphi}\cos\theta) \tag{152}$$

$$= I_1\sin^2\theta\dot{\varphi} + I_3\Omega_3\cos\theta \tag{153}$$

$$E = \frac{1}{2}I_1(\dot{\theta}^2 + \dot{\varphi}^2\sin^2\theta) + \frac{1}{2}I_3\Omega_3^2 + \mu gl\cos\theta \tag{154}$$

are all conserved. Defining the constants

$$a \equiv \frac{P_\psi}{I_1} = \frac{I_3\Omega_3}{I_1}$$

$$b \equiv \frac{P_\varphi}{I_1}$$

This gives

$$\dot{\varphi} = \frac{b - a\cos\theta}{\sin^2\theta} \quad , \quad \dot{\psi} = \frac{I_1 a}{I_3} - \frac{(b - a\cos\theta)\cos\theta}{\sin^2\theta} \tag{155}$$

[Note: if θ is constant, $\dot{\varphi}$ and $\dot{\psi}$ are also constant. The top spins about the $\hat{x_3}$ axis and precesses about the z-axis at constant speeds, as we found for the previous case. In the more general case, $\dot{\varphi}$ and $\dot{\psi}$ change as θ changes.]

Next, note that we can write the equation for conservation of energy in the form

$$E' = \frac{1}{2} I_1 \dot{\theta}^2 + V_{eff}(\theta) \tag{156}$$

where

$$E' = E - \frac{1}{2} I_3^2 \Omega_3^2 = \text{ constant} \tag{157}$$

and

$$V_{eff}(\theta) = \frac{I_1 (b - a\cos\theta)^2}{2\sin^2\theta} + \mu g l \cos\theta \tag{158}$$

Notice the above equation looks like an effective 1D problem for θ!
So from our previous work on 1D problem, we can immediately reduce the problem to quadratures:

$$t = \int \frac{d\theta}{\sqrt{2(E' - V_{eff}(\theta))/I_1}} \tag{159}$$

So if we could do this integral analytically, we'd get $\theta(t)$. Unfortunately, it's an elliptic integral, so let's try to understand the motion more qualitatively, as we did for orbital motion in various potentials.

Qualitative Analysis of Quadratures

Consider the following system:

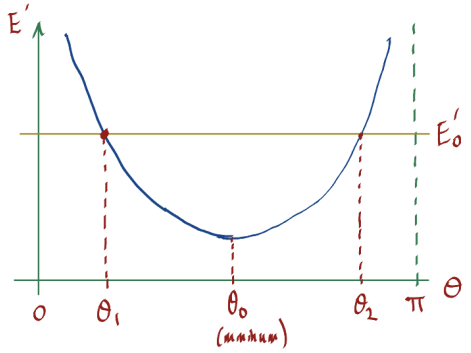


Figure 62: Potential Curve

Such that $V_{eff} \to \infty$ as $\theta \to 0$ and $\theta \to \pi$. Therefore it has the form shown in the figure.

When $E' = E'_0$, θ oscillates between θ_1 & $\theta_2 \to$ nutation!

Special case: at rest at $\theta_0 \to \theta$ is constant during motion $\to$ pure precession (as before)

For general nutation, as θ oscialltes, $\dot{\varphi}$ and $\dot{\psi}$ change (precession rate speeds up & slows down) because

$$\dot{\varphi} = \frac{b - a\cos\theta}{\sin^2\theta} \qquad \dot{\psi} = \frac{I_1 a}{I_3} - \frac{(b - a\cos\theta)\cos\theta}{\sin^2\theta}$$
$$= \frac{P_\varphi - I_3\Omega_3\cos\theta}{I_1\sin^2\theta}$$

Depending on the sign/size of P_φ and $I_3\Omega_3$, $\dot{\varphi}$ can either have the same sign throughout the motion, or it can change sign:

1. no sign change: direction of precession is the same throughout motion, but it speeds up and slows down:

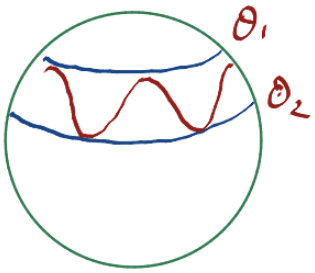


Figure 63: No Sign Change

2. sign change: direction of precession changes during motion:

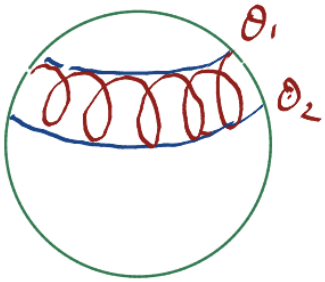


Figure 64: With Sign Change

3. $\dot{\varphi} = 0$ at one extremum of motion: this happens if you hold the tip of the top still while it spins, and then drop it. Since $\theta = \theta_1$, it has the initial condition $\dot{\varphi} = 0$

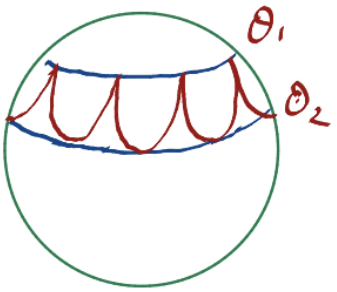


Figure 65: $\dot{\varphi} = 0$

Quantitatively,

$$P_\varphi = I_1 \dot{\varphi} \sin^2\theta + I_3 \Omega_3 \cos\theta = \text{ constant} \tag{160}$$

$$= I_3 \omega_3 \cos\theta_0 \text{ since initially } \dot{\varphi}_0 = 0 \tag{161}$$

Therefore, when θ increases, $\cos\theta$ decreases $\Rightarrow$ $\dot{\varphi}$ increases, starts to precess, which traces out the path shown.

Hamiltonian Formalism

Introduction

Recall Lagrangian: $L(q, \dot{q}, t)$, i=i...n

$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{q}_i}) = \frac{\partial L}{\partial q_i} \tag{162}$$

In the Lagrangian formalism, the state of the system is given by $\{q_i(t), \dot{q}_i(t)\}$: if you know the q_i's and $\dot{q}_i$'s at one time, can predict them at any future time.

In the Hamiltonian formalism, we instead describe the state of the system by $\{q_i(t), p_i(t)\}$

Consider the following function:

$$H(q_i, p_i, t) = \sum_i P_i \dot{q}_i - L(q_i, \dot{q}_i, t) \tag{163}$$

where $P_i \equiv \frac{\partial L}{\partial \dot{q}_i}$ gives $P_i(q_I, \dot{q}_i, t)$...inverting this relation gives $\dot{q}_i(q_i, P_i, t)$ on the RHS, so H is only a function of q_i & P_i (not $\dot{q}_i$).

Now, consider the variation of H under a small change in q_i & P_i:

$$dH = \sum_i (P_i d\dot{q}_i + q_i d\dot{P}_i - \frac{\partial L}{\partial q_i} dq_i - \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i) + \frac{\partial L}{\partial t} dt$$

$\rightarrow$ the terms proportional to $d\dot{q}_i$ cancel, leaving us with

$$dH = \sum_i \left[\dot{q}_i dP_i - \frac{\partial L}{\partial q_i} dq_i \right] - \frac{\partial L}{\partial t} dt$$

$$= \sum_i \left[\dot{q}_i dP_i - \dot{P}_i dq_i \right] - \frac{\partial L}{\partial t} dt$$

But since $H = H(P_i\, q_i\, t)$, on general grounds we know

$$dH = \frac{\partial H}{\partial q_i} dq_i + \frac{\partial H}{\partial P_i} dP_i + \frac{\partial H}{\partial t} dt$$

So we arrive with

$$\dot{q}_i = \frac{\partial H}{\partial P_i} \quad , \quad \dot{P}_i = -\frac{\partial H}{\partial q_i} \quad , \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t} \tag{164}$$

These are known as Hamilton's Equations. They are equivalent to the E-L equations, but instead of n 2nd order PDEs, they are 2n 1st order PDEs. In addition,

1. They are expressed in terms of P_i & q_i, not q_i & $\dot{q}_i$
2. They are nicely symmetric in P_i & q_i, which will turn out to have elegant consequences

"The Canonical Equations"

Example 1 Particle in a Potential

$$l = \frac{1}{2}m\dot{\vec{r}}^2 - V(\vec{r})$$

$$P_i = \frac{\partial L}{\partial \dot{r}_i} = m\dot{r}_i \text{ (the same as the physical momentum in this case)}$$

$$\therefore \dot{r}_i = p_i/m$$

$$H = \vec{P}\cdot\dot{\vec{r}} - L$$

$$\frac{1}{m}\vec{P}^2 - \frac{1}{2}m(\frac{\vec{P}}{m})^2 + V(\vec{r}) \quad = \frac{\vec{P}^2}{2m} + V(\vec{r})$$

and the canonical equations are

$$\dot{r}_i = \frac{\partial H}{\partial P_i} = \frac{P_i}{m} \quad \text{(and so } P_i = m\dot{r}_i \text{)} \tag{165}$$

$$\dot{P}_i = -\frac{\partial H}{\partial r_i} = -\frac{\partial V}{\partial r_i} \Rightarrow \dot{\vec{P}} = -\vec{\nabla}V(\vec{r}) \tag{166}$$

Simple Pendulum

Consider the following system

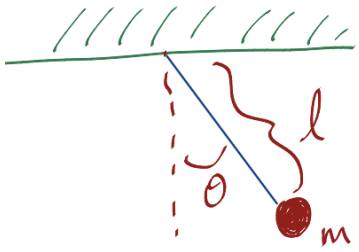


Figure 66: Simple Pendulum

$$L = \frac{1}{2}ml^2\dot{\theta}^2 + mgl\cos\theta \tag{167}$$

$$P_\theta = \frac{\partial L}{\partial \dot{\theta}} = ml^2\dot{\theta}$$

$$H = \frac{P_\theta^2}{sml^2} - mgl\cos\theta$$

$$\Rightarrow \dot{P}_\theta = -\frac{\partial H}{\partial \theta} = -mlg\sin\theta \tag{168}$$

$$\dot{\theta} = \frac{\partial H}{\partial P_\theta} = \frac{P_\theta}{ml^2} \tag{169}$$

[Aside: the transformation from $L(q_i, \dot{q}_i), t$ to $H(q_i, P_i, t)$ is actually a standard piece of mathematics, called a Legendre Transform:

If I have a function $f(x, y)$
The Legendre transform of f is

$$g(u, y) \equiv ux - f(x, y) \text{ where } u(x, y) = \frac{\partial f}{\partial x} \tag{170}$$

is inverted to give $x(u, y)$

So g is a function of u, not x. And $H(P_i, q_i, t)$ is the Legendre transform of $L(q_i, \dot{q}_i, t)$.

In our case,

$$\begin{aligned} f &\leftrightarrow L \\ g &\leftrightarrow H \\ y &\leftrightarrow q \\ x &\leftrightarrow \dot{q} \\ u &\leftrightarrow P \end{aligned} \tag{171}$$

Legendre transform show up in various places in Physics (ex. thermodynamics, quantum field theory), but since we won't need anything beyond what we won't go into more detail here.]

Phase Space Flow

It is convenient to visualize solution of Hamilton's equations by plotting them in 2n-dimensional phase space $\{q_i, P_i\}$. Since $\dot{P}_i = -\frac{\partial H}{\partial q_i}$ & $\dot{q}_i = \frac{\partial H}{\partial P_i}$ are 1st order, given $P_i(0)$ & $q_i(0)$ (or at any other fixed time) we can find $P_i(t)$ & $q_i(t)$ at any time t.

Using simple pendulum as an example, here is its phase space flow:

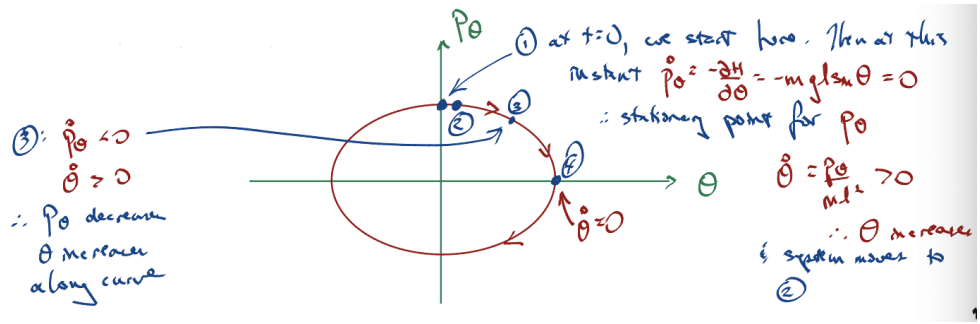


Figure 67: Phase Flow of Simple Pendulum

Note that at each point in phase space, $\{\theta, P_\theta\}$ uniquely determines behaviour of the system $\Rightarrow$ different paths in phase space never cross!

So for example, if we start with different initial conditions (point B), we would get another path in phase space which never intersects the first:

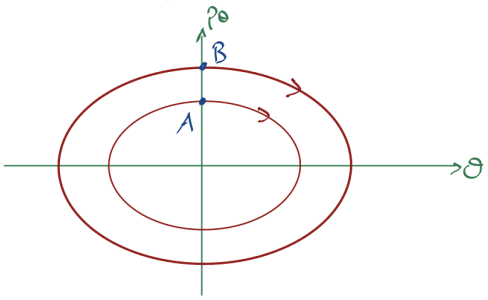


Figure 68: Phase Space with different Initial Condition

Note that for the simple pendulum, the phase space trajectories are only ellipses for small oscillations. More generally, the possible "phase space flows" will look like this:

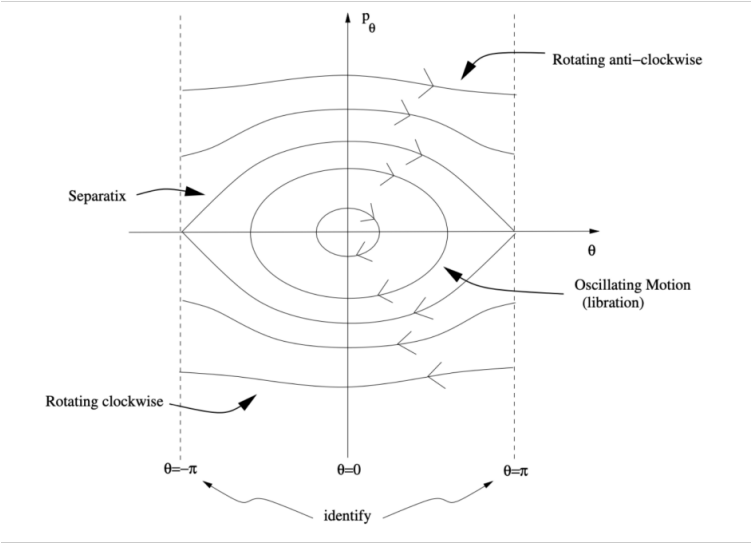


Figure 69: Phase Flow

Phase space flow tells us how the q_i's & P_i's evolve in time. More generally, the physical quantities we are interested in are functions of the q_i's & P_i's. We can use Hamilton's equations to figure out how they evolve along phase space flow.

1. Energy: We know from Noether's theorem that if $\frac{\partial L}{\partial t} = 0$, H is conserved. This falls out trivially from Hamilton's Equations:

$$\begin{aligned}\frac{\partial H}{\partial t}(q_i, P_i, t) &= \frac{\partial H}{\partial q_i}\dot{q}_i + \frac{\partial H}{\partial P_i}\dot{P}_i + \frac{\partial H}{\partial t} \\ &= -\dot{P}_i\dot{q}_i + \dot{q}_i\dot{P}_i = 0\end{aligned}$$

2. Cyclic coordinates: If q_i is cyclic, then

$$\frac{\partial H}{\partial q_i} = 0 \Rightarrow \dot{P}_i = 0 \Rightarrow P_i \text{ is conserved}$$

3. General: Now let's consider an arbitrary function $f(q_i, P_i, t)$ and look how it changes along a trajectory in phase space:

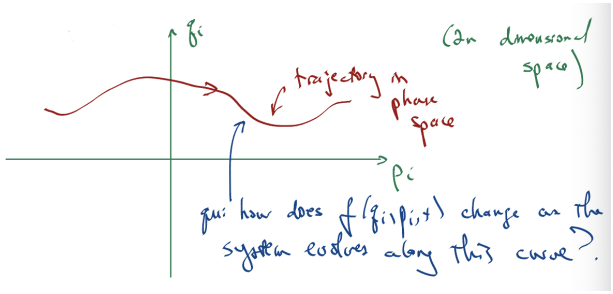


Figure 70: Arbitrary Function in Phase Space

$$\begin{aligned}
\frac{d}{dt}f(q_i, P_i, t) &= \sum_i \left(\frac{\partial f}{\partial P_i}\dot{P}_i + \frac{\partial f}{\partial q_i}\dot{q}_i\right) + \frac{\partial f}{\partial t} \\
&= \sum_i \left[-\frac{\partial f}{\partial P_i}\frac{\partial H}{\partial q_i} + \frac{\partial f}{\partial q_i}\frac{\partial H}{\partial P_i}\right] + \frac{\partial f}{\partial t} \\
&\equiv \{f, H\} + \frac{\partial f}{\partial t} \\
\text{where } \{f, g\} &\equiv \sum_i \left[\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial P_i} - \frac{\partial g}{\partial q_i}\frac{\partial f}{\partial P_i}\right] \qquad (172)
\end{aligned}$$

is called the Poisson Bracket of f & g.

As we will see, Poisson brackets are useful! Here's one way: suppose $f(q_i, P_i)$ has no explicit time dependence, $\frac{\partial f}{\partial t} = 0$, then

$$\frac{df}{dt} = \{f, H\} \Rightarrow \text{ if } f, H = 0, \text{ then} \qquad (173)$$

f is an integral of motion! (This is a very simple way to think about conserved quantities.)

Useful Properties

$$\begin{aligned}
&\{f, g\} = -\{g, f\} && \text{anti-symmetric} \\
&\{\alpha f + \beta g, h\} = \alpha\{f, h\} + \beta\{g, h\} && \text{linear} \\
&\{fg, h\} = f\{g, h\} + \{f, h\}g && \text{Leibniz rule} \\
&\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0 && \text{Jacobi identity}
\end{aligned} \qquad (174)$$

These relations will prove useful to simplify expressions. Note that these relations are also satisfied by the commutator of two matrices/linear operators $[a, b] = ab - ba$

Example 2 Simple Poisson Bracket

$$\{q_i, q_j\} = \sum_k (\frac{\partial q_i}{\partial q_k}\frac{\partial q_j}{\partial p_k} - \frac{\partial q_i}{\partial p_k}\frac{\partial p_j}{\partial q_k}) = 0 \tag{175}$$

&

$$\{p_i, p_j\} = 0 \text{ (by a similar argument)} \tag{176}$$

but

$$\{q_i, p_j\} = \sum_k \left[\frac{\partial q_i}{\partial q_k}\frac{\partial p_j}{\partial p_k} - \frac{\partial q_i}{\partial p_k}\frac{\partial p_j}{\partial q_k}\right] = \delta_{ij} \tag{177}$$

Example 3 Poisson Bracket for Cyclic Coordinates

We have

$$\frac{\partial H}{\partial q_i} = 0$$

$$\{p_i, H\} = \sum_k \left[\frac{\partial p_i}{\partial q_k}\frac{\partial H}{\partial p_k} - \frac{\partial p_i}{\partial p_k}\frac{\partial H}{\partial q_k}\right] = 0 \tag{178}$$

$$\Rightarrow p_i \text{ is conserved}$$

Poisson Bracket for Integrals of Motion

Suppose we have 2 integrals of motion, I & J

$$\{H, I\} = \{H, J\} = 0$$

then by Leibniz rule

$$\begin{aligned} \{H, \{I, J\}\} &= -\{I, \{J, H\}\} - \{J, \{H, I\}\} \\ &= 0 \end{aligned} \tag{179}$$

$\therefore \{I, J\}$ is also a constant of the motion/conserved quantity! (of course if $\{I, J\} = 0$ this is trivially true and we don't get a new conservation law.)

The constants of motion form a "closed algebra" under the Poisson bracket (technically, this is called "Lie Algebra")

For example, consider angular momentum

$$\vec{M} \text{ where } \left[M_i = \sum_{jk} \epsilon_{ijk} r_j p_k\right]$$

$$\text{So } M_1 = r_2 p_3 - r_3 p_2 \text{ , etc.}$$

Then we have

$$\begin{aligned} \{M_1, M_2\} &= \{r_2 p_3 - r_3 p_2, r_3 p_1 - r_1 p_3\} \\ &= \{r_2 p_3, r_3 p_1\} + \{r_3 p_2, r_1 p_3\} - \{r_3 p_2, r_3 p_1\} - \{r_2 p_3, r_1 p_3\} \end{aligned} \tag{180}$$

Using Leibniz rule, we arrive with

$$\{M_1, M_2\} = r_1 p_2 - r_2 p_1 = M_3 \tag{181}$$

Therefore, if M_1 & M_2 are conserved, so is M_3 (We can't have a system where M_1 & M_2 are conserved, but not M_3!)

Exercise: Show that $\{M^2, M_i\} = 0$

Canonical Transformation

In this standard (Lagrangian) formalism, we're used to choosing our generalized coordinate to simplify problems (Cartesian, polar, etc.). But since Hamilton's equations are so symmetrical in p_i's & q_i's, we can consider a much large class of cordinate transformations

$$q + i \to Q_i(q,p) \qquad p_i \to P_i(q,p) \tag{182}$$

For arbitrary functions, Q_i & P_i this transformation isn't useful, became in general the new coordinates & momenta won't obey

$$\dot{Q}_i = \frac{\partial H}{\partial P_i} \quad , \dot{P}_i = -\frac{\partial H}{\partial Q_i} \tag{183}$$

For example, swapping p & q, we have

$$Q = p \quad P = q$$

Then

$$\dot{q} = \frac{\partial H}{\partial p} \quad \dot{p} = -\frac{\partial H}{\partial q}$$
$$\Rightarrow \dot{P}\frac{\partial H}{\partial Q} \quad \dot{Q} = -\frac{\partial H}{\partial P}$$

These are not Hamilton's equations.

The class of transformation for which Q_i & P_i satisfy Hamilton's equations are called Canonical Transformations. Let's work out the requirement for a transformation to be canonical for the case of a single degree of freedom $q(t) \to$ it will be easy to generalize this to n degrees of freedom later on.

$$q, p \Rightarrow \text{ define a new generalized coordinate } Q(q,p)$$

$$\text{and a new generalized momentum } P(q,p)$$

Then

$$\dot{Q} = \frac{\partial Q}{\partial q}\dot{q} + \frac{\partial Q}{\partial p}\dot{p} \qquad \dot{P} = \frac{\partial P}{\partial q}\dot{q} + \frac{\partial P}{\partial p}\dot{p} \tag{184}$$

or, in matrix notation,

$$\begin{bmatrix}\dot{Q}\\ \dot{P}\end{bmatrix} = \begin{bmatrix}\frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p}\\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p}\end{bmatrix}\begin{bmatrix}\dot{q}\\ \dot{p}\end{bmatrix} \equiv g \tag{185}$$

We can also write Hamilton's equations in matrix form:

$$\begin{bmatrix}\dot{q}\\ \dot{p}\end{bmatrix} = \begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}\begin{bmatrix}\partial H/\partial q\\ \partial H/\partial p\end{bmatrix} \equiv J \tag{186}$$

finally, we note that

$$\begin{bmatrix}\partial H/\partial q\\ \partial H/\partial p\end{bmatrix} = \begin{bmatrix}\partial Q/\partial q & \partial P/\partial q\\ \partial Q/\partial p & \partial P \partial p\end{bmatrix}\begin{bmatrix}\partial H/\partial Q\\ \partial H/\partial P\end{bmatrix} \tag{187}$$

$\therefore$ the new variable obey

$$\begin{bmatrix}\dot{Q}\\ \dot{P}\end{bmatrix} = AJA^T \begin{bmatrix}\partial H/\partial Q\\ \partial H/\partial P\end{bmatrix} \tag{188}$$

So, if $AJA^T = J$, this is just Hamilton's equations in the new variables

$$\dot{Q} = \frac{\partial H}{\partial P} \quad \dot{P} = -\frac{\partial H}{\partial Q} \tag{189}$$

and the transformation is canonical.

Poisson Brackets under Canonical Transformation

Consider two functions $f(g,p)$ & $g(f,p)$ We have

$$\begin{aligned}\{f,g\} &= \frac{\partial f}{\partial g}\frac{\partial q}{\partial p} - \frac{\partial f}{\partial p}\frac{\partial g}{\partial q}\\ &= \begin{bmatrix}\frac{\partial f}{\partial q} & \frac{\partial f}{\partial p}\end{bmatrix}\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}\begin{bmatrix}\frac{\partial g}{\partial q}\\ \frac{\partial g}{\partial p}\end{bmatrix}\end{aligned}$$

but we also know

$$\begin{bmatrix}\frac{\partial g}{\partial q}\\ \frac{\partial g}{\partial p}\end{bmatrix} = \begin{bmatrix}\partial Q/\partial q & \partial P/\partial q\\ \partial Q/\partial p & \partial P/\partial p\end{bmatrix}\begin{bmatrix}\partial g/\partial Q\\ \partial g/\partial P\end{bmatrix}$$

(and similarly for f), so we have

$$\begin{aligned}\{f,g\} &= (\frac{\partial f}{\partial Q}\frac{\partial g}{\partial P})AJA^T\begin{bmatrix}\partial g/\partial Q\\ \partial g/\partial P\end{bmatrix}\\ &= \frac{\partial f}{\partial Q}\frac{\partial g}{\partial P} - \frac{\partial f}{\partial P}\frac{\partial g}{\partial Q}\end{aligned} \tag{190}$$

$\therefore$ Poisson brackets are preserved under canonical transformations.
[In particular, $\{Q,Q\} = \{P,P\} = 0$, and $\{Q,P\} = 1$]

All of this is easily extended to n degrees of freedom:

$$q_i \to Q_i(\{g,p\}) \qquad p_i \to P_i(\{g,p\}) \qquad i = 1...n$$

is canonical iff $AJA^T = J$
where now

$$J = \begin{bmatrix}0 & \mathbb{I}_{n\times n}\\ -\mathbb{I}_{n\times n} & 0\end{bmatrix} \tag{191}$$

$$\text{and } A = \begin{bmatrix}\frac{\partial Q_1}{\partial q_1}\frac{\partial Q_1}{\partial q_2} & \cdots & \frac{\partial Q_1}{\partial q_n}\frac{\partial Q_1}{\partial p_1} & \cdots & \frac{\partial Q_1}{\partial p_n}\\ \frac{\partial Q_2}{\partial q_1} & \cdots & & \cdots & \frac{\partial Q_2}{\partial p_n}\\ \vdots & & & & \vdots\\ \frac{\partial P_n}{\partial q_1} & & & & \frac{\partial P_n}{\partial p_n}\end{bmatrix} \tag{192}$$

To check if a transofrmation is canonical, it's sufficient to show either that

$$J = AJA^T \tag{193}$$

or

$$\begin{aligned} \{Q_i, Q_j\} = \{P_i, P_j\} = 0 \\ \{Q_i, P_j\} = \delta_{ij} \end{aligned} \tag{194}$$

Example 4 $Q_i = p_i$ **and** $P_i = -q_i$

This always preserved Poisson brackets:

$$\{P_i, P_j\} = \{Q_i, Q_j\} = 0 \ , \ \{Q_i, P_j\} = -\{p_i, q_j\} = -(-\delta_{ij}) = \delta_{ij}$$

explicitly

$$\begin{aligned} \therefore AJA^T &= \begin{bmatrix} 0 & \mathbb{I} \\ -\mathbb{I} & 0 \end{bmatrix} \begin{bmatrix} 0 & \mathbb{I} \\ -\mathbb{I} & 0 \end{bmatrix} \begin{bmatrix} 0 & -\mathbb{I} \\ \mathbb{I} & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & \mathbb{I} \\ -\mathbb{I} & 0 \end{bmatrix} = J \end{aligned}$$

Therefore, A is symplectic.

Example 5 Simple Harmonic Oscillator

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 q^2 \tag{195}$$

Hamilton:

$$\dot{p} = -\frac{\partial H}{\partial q} = -\omega^2 q \ , \ \dot{q} = p/m \tag{196}$$

$$\begin{aligned} \Rightarrow \ddot{q} = \frac{\dot{p}}{m} = -\omega^2 q \Rightarrow q(t) &= A \sin\left[\omega(t - t_0)\right] \\ \& \quad p(t) &= m\dot{q} = m\omega A\omega \left[\omega(t - t_0)\right] \end{aligned}$$

Phase space:

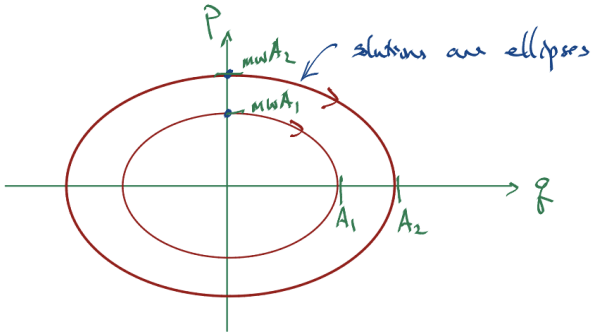


Figure 71: Phase Space

Here's a really simple canonical transformation:

$$Q = \sqrt{m\omega}q \quad P = p/\sqrt{m\omega} \tag{197}$$

in terms which

$$H = \frac{\omega}{2}(P^2 + Q^2) \tag{198}$$

Solutions are

$$\begin{aligned} Q &= A\sin\left[\omega(t - t_0)\right] \\ P &= A\cos\left[\omega(t - t_0)\right] \end{aligned} \tag{199}$$

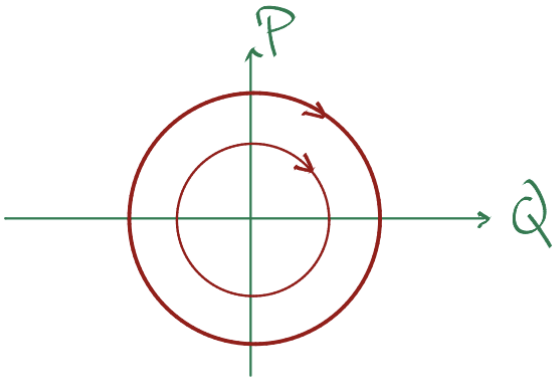


Figure 72: Solutions in Phase Space

So let's do a canonical transformation to "polar" coordinate (not in real space, but in phase space)

$$(\theta, I) \quad \text{where} \quad \begin{aligned} Q &= \sqrt{2I}\sin\theta \\ P &= \sqrt{2I}\cos\theta \end{aligned}$$

or

$$\begin{aligned} I &= \frac{1}{2}(P^2 + Q^2) \\ \theta &= \tan^{-1} P/Q \end{aligned} \tag{200}$$

You can check that this is a canonical transformation. Then we have $H = I\omega$ independent of θ
$\therefore$ θ is cyclic, and its conjugate momentum I is conserved. Explicitly, Hamilton's equations are

$$\dot{\theta} = \frac{\partial H}{\partial I} = \omega, \quad \dot{I} = -\frac{\partial H}{\partial \theta} = 0 \tag{201}$$

and the phase space flow is trivial:

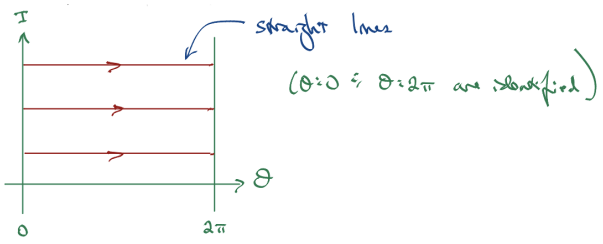


Figure 73: Phase Space Flow

(θ, I) are called "action-angle variables", and this construction can be extended to more complex systems.

For more complex system, you can convert to action-angle variable by finding a canonical transformation

$$(q_i, p_i) \to (Q_i, I_i) \tag{202}$$

such that $H = H(I_1, ..., I_n)$ doesn't depend on Q_i's (cyclic). Then $\dot{I}_i = -\frac{\partial H}{\partial \theta_i} = 0 \Rightarrow I_i$'s are conserved, and

$$\dot{\theta}_i = \frac{\partial H}{\partial I_i} = \omega_i(I_1, ..., I_n) \Rightarrow Q_i = \omega_i t \tag{203}$$

$\Rightarrow$ EOM are very simple in these coordinates.

Of course, for a general system, you need n conserved quantities $\Rightarrow$ integrable system!

Theorem (Liouville-Arnold): any system undergoing bounded motion which has n mutually Poisson-commuting integrable of motion $I_1 ... I_n$, $\{I_i, I_j\} = 0$ can be written in action-angle variable & is integrable.

Example 6 Particle in Central Potential

This system has $n = 3$ degrees of freedom. We found 4 (general potential) or 5 (1/r potential) integrable of motion $H, \vec{M}$ (& $\vec{A}$) but they are not all mutually Poisson commuting:

$$\{M_i, M_j\} = \epsilon_{ijk} M_k \tag{204}$$

but H, M^2 & M_z all Poisson commute. Therefore, system have n mutually commuting integrable of motion, and can be written in action-angle variables.

$$\begin{aligned} I_\varphi &= P_\varphi = mr^2\dot{\varphi} \\ I_r &= \sqrt{\frac{m}{2|E|}}k - I_\varphi \end{aligned} \tag{205}$$

$$H(I_r, I_\varphi) = -\frac{mk^2}{2(I_r + I_\varphi)^2} \tag{206}$$

Infinitesimal canonical transformations & Noether's theorem

Consider an infinitesimal transformation

$$\begin{aligned} q_i \to Q_i &= q_i + \alpha F_i(q,p) \\ p_i \to P_i &= p_i + \alpha E_i(q,p) \end{aligned} \tag{207}$$

where α is infinitesimal. This gives

$$\begin{aligned} \{Q_i, P_j\} &= \{q_i, p_j\} + \alpha\left[\{F_i, p_j\} + \{q_i, E_j\}\right] + O(\alpha^2) \\ &= \delta_{ij} + \alpha \sum_k \left[\frac{\partial F_i}{\partial q_k}\frac{\partial p_j}{\partial p_k} - \frac{\partial F_i}{\partial p_k}\frac{\partial p_j}{\partial q_k} + \frac{\partial q_i}{\partial q_k}\frac{\partial E_j}{\partial p_k} - \frac{\partial q_i}{\partial p_k}\frac{\partial E_j}{\partial q_k} + ...\right] \\ &= \delta_{ij} + \alpha\left[\frac{\partial F_i}{\partial q_j} + \frac{\partial E_j}{\partial p_k}\right] + O(\alpha^2) \end{aligned} \tag{208}$$

Similarly, check

$$\begin{aligned} \{Q_i, Q_j\} &= \alpha\left[\frac{\partial F_j}{\partial p_i} - \frac{\partial F_i}{\partial p_j}\right] \\ \{P_i, P_j\} &= \alpha\left[\frac{\partial E_i}{\partial q_j} - \frac{\partial E_j}{\partial q_i}\right] \end{aligned} \tag{209}$$

Therefore for a transformation to be canonical, we must have

$$\frac{\partial F_i}{\partial p_j} = \frac{\partial F_j}{\partial p_i} \quad , \quad \frac{\partial E_i}{\partial q_j} = \frac{\partial E_j}{\partial q_i} \quad \text{and} \quad \frac{\partial F_i}{\partial q_j} = -\frac{\partial E_j}{\partial p_i} \tag{210}$$

This is true if $F_i = \frac{\partial G}{\partial p_i}$, $E_i = -\frac{\partial G}{\partial q_i}$ for some function $G(q_i, p_i)$, which is called generator of the transformation.

Example 7 Physical significance of Generator function G

When $G = P_k$, k^{th} component of momentum

$$\begin{aligned} F_i = \delta_{ik} \ , \ E_i = 0 \Rightarrow q_i &\to q_i + \alpha\delta_{ik} \\ p_i &\to p_i \end{aligned}$$

When $G = M_z$, z component of angular momentum. We already know that in spherical polar coordinates, M_z is the momentum conjugate to φ: $p_\varphi = M_z$

$$\therefore F_i = \frac{\partial G}{\partial p_i} = \delta_{i\varphi} \quad , E_i = 0$$

Therefore, the canonical transformation generated by $G = M_z$ is

$$\varphi \to \varphi + \alpha \quad r, \theta, p_i \text{ unchanged}$$

When $G = H$, we have

$$\begin{aligned} \frac{dq_i}{d\alpha} &= \frac{\partial H}{\partial p_i} \\ \frac{\partial p_i}{\partial \alpha} &= -\frac{\partial H}{\partial q_i} \end{aligned}$$

Hamilton's equations for a translation in time:

$$q_i(t) \to q_i(t+\alpha)$$
$$p_i(t) \to p_i(t+\alpha)$$

Recall we know from Noether's theorem:

$$\begin{aligned} q_k \to q_k + \alpha \Rightarrow p_k &\quad \text{conserved} \\ \varphi \to \varphi + \alpha \Rightarrow M_z &\quad \text{conserved} \\ t \to t + \alpha \Rightarrow H &\quad \text{conserved} \end{aligned} \tag{211}$$

Now we see that in each case, the conserved quantity generates the corresponding canonical transformation! This suggests a correspondance between Noether's theorem and canonical transformations...

Up to now, we've thought of canonical transformations as a change of variables we use to describe the same point in phase space ("passive" transformation).

Example 8 Active Interpretation

We have $(Q, P) = (q + \alpha, p)$

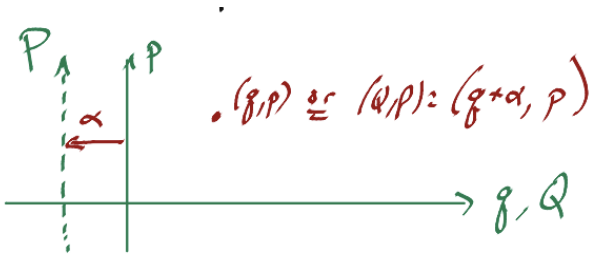


Figure 74: Passive Transformation

but we could equally well consider the "active" interpretation that a canonical transformation takes us from one point (q, p) to another $(q, p) = (Q(\alpha), P(\alpha))$ in phase space:

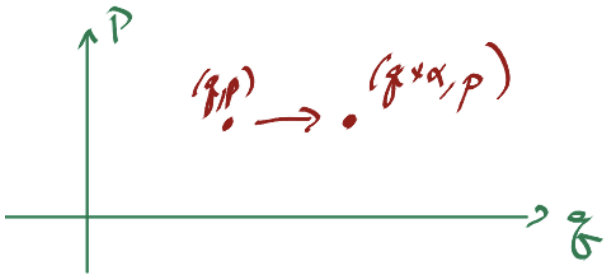


Figure 75: Active Transformation

So as we vary α, our transformation traces out a curve in phase space

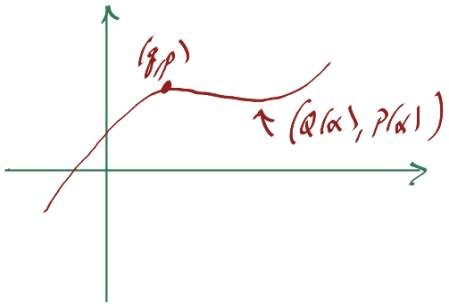


Figure 76: Phase Curve

Note that this curve does not in general correspond to time evaluation of the system, but instead the transformation parametrized by $(Q(\alpha), P(\alpha))$. This transformation is a symmetry if it leaves H unchanged (Hamilton's equations are unchanged under the transformation)

$$\begin{aligned} \therefore \delta H &= \frac{\partial H}{\partial q_i}\delta q_i + \frac{\partial H}{\partial p_i}\delta p_i = 0 \\ &= \alpha\frac{\partial H}{\partial q_i}\frac{\partial G}{\partial p_i} - \alpha\frac{\partial H}{\partial p_i}\frac{\partial G}{\partial q_i} + O(\alpha^2) \\ &= \alpha\{G, H\} = 0 \end{aligned} \tag{212}$$

$\therefore$ If G is conserved, it generates an active canonical transformation which is a symmetry (!) (as in our earlier examples, but now shown to be true in general.)

This is Noether's theorem in the Hamiltonian formalism $\rightarrow$ very simple! Conserved quantities generate symmetries of the system.

Different Perspective on Transformation

Consider the transformation

$$\begin{aligned} q_i &\rightarrow Q_i(p, q, \alpha) \\ p_i &\rightarrow P_i(p, q, \alpha) \end{aligned} \tag{213}$$

Then

$$\begin{aligned} \frac{dq_i}{d\alpha} &= F_i(p, q) = \frac{\partial G}{\partial p_i} \\ \frac{dp_i}{d\alpha} &= E_i(p, q) = -\frac{\partial G}{\partial q_i} \end{aligned} \tag{214}$$

So, if $G = H$, it generates time translation.
and if $G = p_x$, it generates spatial translation.
and if $G = M_z$, it generates rotations about z-axis.

$\therefore$ a one parameter symmetry transformation = "Hamiltonian flow" on phase space, where the "Hamiltonian" is the corresponding conserved quantity!

Finally, if we have a function $f(q_i, p_i)$, how does it change under the transformation $(q_i, p_i) \to (Q_i(\alpha), P_i(\alpha))$?

$$\frac{df}{d\alpha} = \frac{\partial f}{\partial p_i}\frac{dp_i}{d\alpha} + \frac{\partial f}{\partial q_i}\frac{dq_i}{d\alpha} = \{f, G\} \tag{215}$$

Therefore, under an infinitesimal transformation $alpha$, f changes by

$$f \to f + \{f, G\}\alpha$$

So if f Poisson commutes with G, it is invariant under the transformation generated by G:

$$\begin{aligned} &\{f, H\} = 0 \text{ , f is time independent} \\ &\{f, p_x\} = 0 \text{ , f is invariant under spatial translation in x direction} \\ &\{f, M_z\} = 0 \text{ , f is invariant under rotations about z axis} \end{aligned} \tag{216}$$

and all of this translates to quantum mechanics with $\{f, g\} \to \frac{-i}{\hbar}\left[\hat{f}, \hat{g}\right]$.